Big Ideas Math Middle School: Hands-On Block & SC-6 Supplement

Section-by-Section Textbook Methods, Step-by-Step Manipulative Block Instructions, Worked Examples & 450 Practice Problems

1. BIG IDEAS TEXTBOOK METHODS

Every section aligns directly to Big Ideas Math (Units 1–10) and tracks the exact textbook methods: Factor Trees, Tape Diagrams, Ratio Tables, LCD Towers, Area Models, Balance Scales, Nets, and Box Plots.

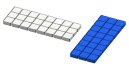
2. STEP-BY-STEP BLOCK KITS

Each section names the exact StackCalc Block Kit used and walks through Step 1, Step 2, and Step 3 with physical blocks alongside 1 clear Worked Example showing both the textbook method and SC-6 keystrokes.

3. WORKSPACE & +10 SCOREBOARD

Every problem gives open workspace to work out the math and write the final answer—never the calculator stack! Optional QUEST-01 Scoreboard: 1 block = +10 pts (1st Way), 2 blocks = +20 pts (2nd Way).

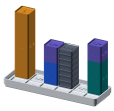
MANIPULATIVE BLOCK KITS & THE 3x1x7U SC-6 SCOREBOARD KIT (QUEST-01)



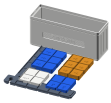
COUNT-01: Array Tiles



PRIME-01: Prime Towers



FRAC-01: 84mm Fractions



ALGB-01: Algebra Stage



TIME-01: Geared Clock

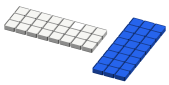
UNIT / CHAPTER & BIG IDEAS ALIGNMENT	SECTIONS & WORKBOOK PAGES	BIG IDEAS TEXTBOOK METHODS
Unit 1: Numerical Expressions & Factors Big Ideas Math Grade 6 — Chapter 1 • Counting & Array Studio (COUNT-01) + Prime Factor Studio (PRIME-01)	[ ] Sec 1.1 (p.2): Whole Number Operations & Division with Remainders [ ] Sec 1.2 (p.3): Powers & Exponents — Squares, Cubes & Growth [ ] Sec 1.3 (p.4): Order of Operations — Grouping Symbols, Exponents & PEMDAS [ ] Sec 1.4 (p.5): Prime Factorization — Factor Trees & Logarithmic Prime Towers [ ] Sec 1.5 (p.6): Greatest Common Factor (GCF) & Least Common Multiple (LCM)	**Arrays, Factor Trees & Venn GCF/LCM** Kit: Counting & Array Studio (COUNT-01) **[ ] Unit 1 Complete (50/50)**
Unit 2: Fractions & Decimals Big Ideas Math Grade 6 — Chapter 2 • Fraction Studio (FRAC-01) + Place Value Studio (PLACE-01)	[ ] Sec 2.1 (p.7): Equivalent Fractions, Mixed Numbers & Adding Towers [ ] Sec 2.2 (p.8): Multiplying Fractions & Mixed Numbers [ ] Sec 2.3 (p.9): Dividing Fractions & Mixed Numbers — Why Reciprocal Flip Works [ ] Sec 2.4 (p.10): Adding, Subtracting & Multiplying Decimals [ ] Sec 2.5 (p.11): Dividing Decimals & The 3-Way Fraction-Decimal-Remainder Tr..	**LCD Towers, Reciprocals & Place Value** Kit: Fraction Studio (FRAC-01) **[ ] Unit 2 Complete (50/50)**
Unit 3: Ratios, Rates & Percents Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Counting & Array Tiles (COUNT-01) + Fraction Studio (FRAC-01)	[ ] Sec 3.1 (p.12): Ratios & Tape Diagrams — Part-to-Part vs. Part-to-Whole [ ] Sec 3.2 (p.13): Ratio Tables & Double Number Lines via [g] [REP] [ ] Sec 3.3 (p.14): Unit Rates & Best-Buy Showdowns (Plus [x↔y] Swap Rescue) [ ] Sec 3.4 (p.15): Percents, Fractions & Decimals — The 3-Costume Switcher [ ] Sec 3.5 (p.16): Discounts, Percent Change ([g] [%CHG]) & Unit Conversions	**Tape Diagrams, Ratio Tables & Unit Rates** Kit: Counting & Array Tiles (COUNT-01) **[ ] Unit 3 Complete (50/50)**
Unit 4: Algebraic Expressions & Properties Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Algebra Boxes & Factoring Stage (ALGB-01) + Counting Tiles (COUNT-01)	[ ] Sec 4.1 (p.17): Evaluating Algebraic Expressions — Variable Boxes on the Stack [ ] Sec 4.2 (p.18): Combining Like Terms — Why x² Flats, x Rods & 1 Cubes Never.. [ ] Sec 4.3 (p.19): The Distributive Property — One Split Rectangle, Two Ways [ ] Sec 4.4 (p.20): Factoring Expressions with GCF — Reverse the Rectangle	**100-Grids, Double Number Lines & % Eq.** Kit: Algebra Boxes & Factoring Stage (A.. **[ ] Unit 4 Complete (50/50)**
Unit 5: Areas of Polygons & Elapsed Time Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Counting & Array Tiles (COUNT-01) + Planetary Geared Clock (TIME-01)	[ ] Sec 5.1 (p.21): Areas of Parallelograms & Rectangles ([f] [RECT]) [ ] Sec 5.2 (p.22): Areas of Triangles — Half a Rectangle ([g] [TRI]) [ ] Sec 5.3 (p.23): Trapezoids & Composite Figures with the [Σ+] Running Ledger [ ] Sec 5.4 (p.24): Coordinate Polygons & Planetary Clock Base-60 Time ([CLK])	**Substitution, Area Models & Like Terms** Kit: Counting & Array Tiles (COUNT-01) **[ ] Unit 5 Complete (50/50)**
Unit 6: Integers, Rational Numbers & The Coordinate Plane Big Ideas Math Grade 6 — Chapter 6 • Ten-Frame Bicolor Counters (BASE-10) + Number Rack (RACK-01)	[ ] Sec 6.1 (p.25): Positive & Negative Integers — Zero Pairs & The [+/-] Sign Flipper [ ] Sec 6.2 (p.26): Comparing & Ordering Integers and Rational Numbers [ ] Sec 6.3 (p.27): Absolute Value — Distance From Zero Is Always Positive ([g] [|x|]) [ ] Sec 6.4 (p.28): The 4-Quadrant Coordinate Plane & Axis Reflections [ ] Sec 6.5 (p.29): Distance Between Points on the Coordinate Plane ([g] [|x|])	**Number Lines, Opposites, |x| & 4-Quad** Kit: Ten-Frame Bicolor Counters (BASE-10) **[ ] Unit 6 Complete (50/50)**
Unit 7: Equations & Inequalities Big Ideas Math Grade 6 — Chapter 7 • Algebra Boxes & Stage (ALGB-01) + Number Rack (RACK-01)	[ ] Sec 7.1 (p.30): Writing Equations & Testing Solutions — The Balance Scale Rule [ ] Sec 7.2 (p.31): Solving One-Step Addition & Subtraction Equations ([f] [LASTX]) [ ] Sec 7.3 (p.32): Solving One-Step Multiplication & Division Equations [ ] Sec 7.4 (p.33): Writing, Graphing & Solving Inequalities ([f] [>], [g] [<], [f] [≥], [g] [.. [ ] Sec 7.5 (p.34): Two-Variable Relationships (y = kx and y = x + b) — Tables & Gr..	**Balance Scale Inverse Ops & Inequalities** Kit: Algebra Boxes & Stage (ALGB-01) **[ ] Unit 7 Complete (50/50)**
Unit 8: Surface Area & Volume of 3D Solids Big Ideas Math Grade 6 — Chapter 8 • Counting & Array Tiles (COUNT-01) + Gridfinity 3x1 Storage Boxes	[ ] Sec 8.1 (p.35): 3D Solids, Nets & Surface Area of Rectangular Prisms ([f] [RECT.. [ ] Sec 8.2 (p.36): Surface Area of Triangular Prisms & Pyramids ([f] [RECT] + [g] [.. [ ] Sec 8.3 (p.37): Volume of Rectangular Prisms with Whole & Fractional Edges ([f.. [ ] Sec 8.4 (p.38): Circles, Cones & Right-Triangle Hypotenuse ([π], [f] [CIRC], [g] [..	**Cut-and-Slide Polygons, 2D Nets & V=Bh** Kit: Counting & Array Tiles (COUNT-01) **[ ] Unit 8 Complete (50/50)**
Unit 9: Statistical Measures & Data Displays Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Counting & Array Tiles (COUNT-01) + Probability Dice Studio	[ ] Sec 9.1 (p.39): Dot Plots, Frequency & Mean as 'Fair Share Leveling' ([Σ+], [g] [.. [ ] Sec 9.2 (p.40): Median, Mode & Range — One Data Entry, Five Instant Answers! [ ] Sec 9.3 (p.41): Quartiles, Interquartile Range ([IQR]) & Box-and-Whisker Plots [ ] Sec 9.4 (p.42): Mean Absolute Deviation ([g] [MAD]) Without the 20-Minute Nigh..	**Dot Plots, Fair-Share Mean, IQR & Box Plot** Kit: Counting & Array Tiles (COUNT-01) **[ ] Unit 9 Complete (50/50)**
Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equat.. Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Ten-Frame Zero-Pair Counters (BASE-10) + Algebra Stage (ALGB-01) + Probability	[ ] Sec 10.1 (p.43): Adding & Subtracting Signed Numbers — Why 'Minus a Negati.. [ ] Sec 10.2 (p.44): Multiplying & Dividing Signed Rationals — The Light-Switch Flip.. [ ] Sec 10.3 (p.45): Solving Two-Step Equations (ax + b = c) — Unwrapping in Rev.. [ ] Sec 10.4 (p.46): Probability, Combinatorics & Graduation Quiz Sprint ([DICE], [x!..	**Zero-Pair Chips, 2-Step Eq., π & Prob.** Kit: Ten-Frame Zero-Pair Counters (BASE.. **[ ] Unit 10 Complete (50/50)**

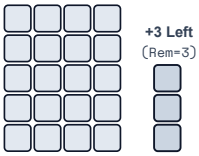
Section 1.1: Whole Number Operations & Division with Remainders

Big Ideas Method: Rectangular Array Model & Division Check (Dividend = Divisor × Quotient + Remainder) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
When you divide real items (like 23 tiles into groups of 4), a decimal like 5.75 hides the actual leftover pieces. Grouping physical Counting Tiles shows the full rows (Quotient) and loose tiles (Remainder), while [g] [÷R] puts Quotient on Floor 1 (X) and Remainder on Floor 2 (Y) at the same time!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and count out 23 unit tiles onto the Gridfinity baseplate.
Step 2: Arrange the tiles into equal rows of 4. Count 5 complete rows of 4 tiles (5 × 4 = 20 tiles).
Step 3: Place the 3 leftover tiles beside the array as Remainder R = 3, proving 23 = 4 × 5 + 3.

WORKED EXAMPLE — EXAMPLE 1: QUOTIENT & REMAINDER WITH BLOCKS + [G] [÷R]

Problem: Divide 23 ÷ 4 and find the quotient and remainder.

Big Ideas Textbook Method (Rectangular Array Model & Division Check (Dividend = Divisor × Quotient..
1. Divide 23 ÷ 4 to find the greatest whole number of groups of 4: 5 groups (5 × 4 = 20).
2. Subtract 23 - 20 = 3 to find the remainder (R = 3, which is less than the divisor 4).
3. Check with the Division Algorithm: Quotient × Divisor + Remainder = 5 × 4 + 3 = 23.

Block Connection (COUNT-01): Snap 23 Counting Tiles into rows of 4. You build 5 full rows (20 tiles) with 3 leftover tiles.

SC-6 Keystrokes: `23` `ENTER` `4` `g` `÷R`

Answer: 5 R 3

Floor 1 (X) shows 5 full rows; Floor 2 (Y) shows 3 left..

SECTION 1.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Rectangular Array Model & Division Check (Dividend = Divisor ×..

1 Big Ideas Method: Equal-Rows Array
Arrange 31 tiles into rows of 6. Find Quotient & Remainder.

Work:

Answer: _____

2 Big Ideas Method: Equal-Rows Array
Divide 47 ÷ 5 using tiles or [g] [÷R].

Work:

Answer: _____

3 Big Ideas Method: Equal-Rows Array
Divide 58 ÷ 7 and check with [f] [LASTX] [×] [+].

Work:

Answer: _____

4 Big Ideas Method: Equal-Rows Array
Divide 85 ÷ 9. How many full groups of 9 and how many left?

Work:

Answer: _____

5 Big Ideas Method: Quotient & Remainder
Divide 146 ÷ 12. Find the Quotient and Remainder.

Work:

Answer: _____

6 Big Ideas Method: Quotient & Remainder
Divide 250 ÷ 36 and prove with [f] [LASTX] [×] [+].

Work:

Answer: _____

7 Big Ideas Method: Quotient & Remainder
Find the missing dividend: Quotient = 14, Divisor = 8, Remainder = 5.

Work:

Answer: _____

8 Big Ideas Method: Quotient & Remainder
Spot the Glitch: Sam says 75 ÷ 8 = 8 R 11. Why is R=11 impossible? Fix it!

Work:

Answer: _____

9 Application & 2-Way Check: Division Check (q × d + r)
A 3D spool has 300g of PLA. Each box takes 42g. How many full boxes can you print, and how much PLA is left?

Work:

Answer: _____

10 Application & 2-Way Check: Division Check (q × d + r)
115 students ride vans that hold 14 kids each. How many vans must be booked so nobody is left behind?

Work:

Answer: _____

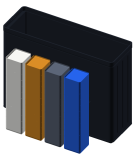
Section 1.2: Powers & Exponents — Squares, Cubes & Growth

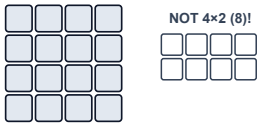
Big Ideas Method: Repeated Multiplication Factors, Perfect Squares (2D Array) & Cubes (3D Stack) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

An exponent tells how many times to use the base as a factor in repeated multiplication. A 2nd power (4^2) builds a 2D square array (4×4 = 16 tiles), and a 3rd power (4^3) stacks 4 square layers into a 3D cube (64 tiles). Evaluate powers using repeated multiplication or [x²], [g] [x³], and [g] [y^x]!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and lay down a row of 4 tiles for the base (4).

Step 2: For 4^2, build a 4 × 4 square array (4 rows of 4 = 16 tiles)—compare to 2 rows of 4 (4 × 2 = 8 tiles).

Step 3: For 4^3, stack 4 identical 4 × 4 square layers high to build a 3D cube of 16 × 4 = 64 tiles.

WORKED EXAMPLE — EXAMPLE 1: SQUARES & CUBES WITH TILES AND [X²] / [G] [X³]

Problem: Write 4^2 and 4^3 as repeated multiplication and find their values. Compare 4^2 to 4 × 2.

Big Ideas Textbook Method (Repeated Multiplication Factors, Perfect Squares (2D Array) & Cubes (3D..

1. Identify the base (4) and the exponent (2 or 3).
2. Write the power as repeated multiplication of the base: $4^2 = 4 \times 4 = 16$.
3. For 4^3, multiply the base 3 times: $4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$ (never multiply base × exponent!).

Block Connection (COUNT-01): Build a 4×4 square of tiles (16 tiles). Stack 4 layers high for 4^3 (64 tiles). Notice 4×2 is only 2 rows of 4 (8 tiles)!

SC-6 Keystrokes: `4` `x²` `4` `g` `x³`

Answer: $4^2 = 16$, $4^3 = 64$

[x²] squares the base; [g] [x³] cubes the base!

SECTION 1.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Repeated Multiplication Factors, Perfect Squares (2D Array) & C..

1 Big Ideas Method: Repeated Factors (b × b)

Write 5^2 as repeated multiplication and find how many tiles are in a 5 × 5 square.

Work:

Answer: _____

2 Big Ideas Method: Repeated Factors (b × b)

Evaluate 7^2 and compare it to 7 × 2.

Work:

Answer: _____

3 Big Ideas Method: Repeated Factors (b × b)

Evaluate 3^3 (a 3 × 3 × 3 cube of blocks).

Work:

Answer: _____

4 Big Ideas Method: Repeated Factors (b × b)

Evaluate 5^3 using repeated multiplication or [g] [x³].

Work:

Answer: _____

5 Big Ideas Method: Square & Cube Evaluation

Evaluate 2^5 by writing the repeated factors and checking with 2 [ENTER] 5 [g] [y^x].

Work:

Answer: _____

6 Big Ideas Method: Square & Cube Evaluation

Evaluate 6^4 using repeated multiplication or [g] [y^x].

Work:

Answer: _____

7 Big Ideas Method: Square & Cube Evaluation

Evaluate 10^4 and 1^4. How many zeros are in 10^4?

Work:

Answer: _____

8 Big Ideas Method: Square & Cube Evaluation

Spot the Glitch: Mia wrote 2^4 = 8 because 2 × 4 = 8. Find the true value of 2^4.

Work:

Answer: _____

9 Application & 2-Way Check: Exponential Growth Model

A cell doubles every hour (2^n). How many cells are there after 6 hours (2^6)? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Exponential Growth Model

Mystery Cube: A giant cube uses 216 unit blocks. How many blocks wide is 1 edge? (Try [g] [x√y]!)

Work:

Answer: _____

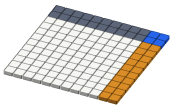
Section 1.3: Order of Operations — Grouping Symbols, Exponents & PEMDAS

Big Ideas Method: Order of Operations (PEMDAS: Grouping Symbols -> Powers -> Multiply/Divide L-to-R -> Add/Subtr.. Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Big Ideas Order of Operations (PEMDAS): (1) Evaluate expressions inside grouping symbols (), [], or fraction bars first; (2) Evaluate powers; (3) Multiply and divide from left to right; (4) Add and subtract from left to right. Grouping tiles on your desk lets you solve each part step by step!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)



GROUP 1 TILES:
8 + 4 = 12

GROUP 2 TILES:
9 - 3 = 6

==> 72

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and use two zones of the lid for the two parentheses groups.

Step 2: In Zone 1, combine 8 tiles and 4 tiles to model (8 + 4) = 12 tiles.

Step 3: In Zone 2, start with 9 tiles and remove 3 tiles to model (9 - 3) = 6, then multiply 12 groups of 6 = 72.

WORKED EXAMPLE — EXAMPLE 1: GROUPING SYMBOLS & ORDER OF OPERATIONS

Problem: Evaluate (8 + 4) × (9 - 3) using the Big Ideas Order of Operations.

Big Ideas Textbook Method (Order of Operations (PEMDAS: Grouping Symbols -> Powers -> Multiply/..

1. Grouping Symbols First: Evaluate inside both sets of parentheses: (8 + 4) = 12 and (9 - 3) = 6.
2. Substitute the simplified values back into the expression: 12 × 6.
3. Multiply to find the final value: 12 × 6 = 72.

Block Connection (COUNT-01): Group 1 has 8 + 4 = 12 tiles. Group 2 has 9 - 3 = 6 tiles. Multiply the two groups: 12 × 6 = 72!

SC-6 Keystrokes: 8 ENTER 4 + 9 ENTER 3 - ×

Answer: 72 *(8 + 4) × (9 - 3) = 12 × 6 = 72!*

SECTION 1.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Order of Operations (PEMDAS: Grouping Symbols -> Powers -> Mult..

1 Big Ideas Method: Grouping Symbols First

Evaluate 5 + 2 × 3. Multiply before adding!

Work:

Answer: _____

2 Big Ideas Method: Grouping Symbols First

Evaluate (14 + 6) ÷ (7 - 2). Solve both parentheses first!

Work:

Answer: _____

3 Big Ideas Method: Grouping Symbols First

Evaluate 30 - 4 × 6.

Work:

Answer: _____

4 Big Ideas Method: Grouping Symbols First

Evaluate (9 + 3) × (15 - 8).

Work:

Answer: _____

5 Big Ideas Method: PEMDAS Step-by-Step

Evaluate 50 - (3 + 4)². Square the sum before subtracting from 50!

Work:

Answer: _____

6 Big Ideas Method: PEMDAS Step-by-Step

Evaluate 120 - 4 × (3 + 2)².

Work:

Answer: _____

7 Big Ideas Method: PEMDAS Step-by-Step

Fraction Bar Grouping: Evaluate (24 + 12) / (9 - 3).

Work:

Answer: _____

8 Big Ideas Method: PEMDAS Step-by-Step

Evaluate 3^3 + 15 ÷ 3 - 2 × 4.

Work:

Answer: _____

9 Application & 2-Way Check: Multi-Step Expression

Evaluate 10 + (15 - 3)² ÷ 9 using PEMDAS step-by-step AND check on your SC-6.

Work:

Answer: _____

10 Application & 2-Way Check: Multi-Step Expression

Movie tickets: 2 adults at $12 each and 4 kids at $8 each, minus a $10 coupon. Write an expression and find total cost.

Work:

Answer: _____

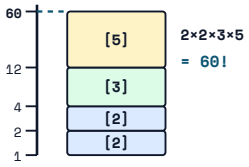
Section 1.4: Prime Factorization — Factor Trees & Logarithmic Prime Towers

Big Ideas Method: Factor Trees, Prime vs. Composite Testing & Prime-Power Factorization Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
In Big Ideas Math, a Factor Tree splits a composite number until every branch ends at a prime number. In the Prime Factor Studio (PRIME-01), block heights are logarithmic (h = 7·log2(p) mm), so [2]+[2]+[3]+[5] always reaches the exact 60 mark whether you split 60 as 6×10 or 4×15!

BLOCK KIT & STEP-BY-STEP GUIDE (PRIME-01: PRIME FACTOR STUDIO)





HOW TO BUILD IT WITH PRIME-01 (PRIME FACTOR STUDIO):

Step 1: Open Kit PRIME-01 (Prime Factor Studio) and set the logarithmic slide-rule scale on the box wall.

Step 2: Select the prime blocks that multiply to 60: two blue [2] blocks, one green [3] block, and one gold [5] block.

Step 3: Stack [2] + [2] + [3] + [5] in any order—the tower height always aligns with 60 on the logarithmic scale!

WORKED EXAMPLE — EXAMPLE 1: FACTOR TREE & PRIME TOWER FOR 60

Problem: Write the prime factorization of 60 using a factor tree, Prime Blocks, and [g] [FACT].

Big Ideas Textbook Method (Factor Trees, Prime vs. Composite Testing & Prime-Power Factorization):

1. Branch 60 into any factor pair (such as 6 × 10 or 4 × 15) on a Factor Tree.
2. Continue splitting composite branches until only prime leaves remain: 6 = 2 × 3 and 10 = 2 × 5.
3. Group identical prime factors using exponents from least to greatest: 60 = $2^2 \times 3 \times 5$.

Block Connection (PRIME-01): Stack two [2] blocks, one [3] block, and one [5] block on the lid—their combined height aligns with 60 on the box wall!

SC-6 Keystrokes: `60` `g` `FACT`

Answer: 60 = $2^2 \times 3 \times 5$ *[PRIME] tests primality; [g] [FACT] ([g][PRIME]) show..*

SECTION 1.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Factor Trees, Prime vs. Composite Testing & Prime-Power Factori..

1 Big Ideas Method: Factor Tree Branches
Draw a factor tree or stack [2] + [2] + [3]. Write the prime factorization of 12.

Work:

Answer: _____

2 Big Ideas Method: Factor Tree Branches
Find the prime factorization of 36 using blocks and [g] [FACT].

Work:

Answer: _____

3 Big Ideas Method: Factor Tree Branches
Test 29 and 57 with [PRIME]. Which one is prime and which is composite?

Work:

Answer: _____

4 Big Ideas Method: Factor Tree Branches
Find the prime factorization of 45.

Work:

Answer: _____

5 Big Ideas Method: Prime-Power Form
Build the Prime Tower for 72. Show how 8 × 9 and 6 × 12 use the exact same blocks.

Work:

Answer: _____

6 Big Ideas Method: Prime-Power Form
Find the prime factorization of 84.

Work:

Answer: _____

7 Big Ideas Method: Prime-Power Form
Find the prime factorization of 100.

Work:

Answer: _____

8 Big Ideas Method: Prime-Power Form
What number has the prime tower [2] + [2] + [3] + [7] ($2^2 \times 3 \times 7$)?

Work:

Answer: _____

9 Application & 2-Way Check: Prime Factorization Check
Find the prime factorization of 180 both by splitting 18 × 10 AND with [g] [FACT].

Work:

Answer: _____

10 Application & 2-Way Check: Prime Factorization Check
Mystery Vault Code: A 3-digit number equals $2^3 \times 3^2 \times 5$. Find the number and verify with [g] [FACT].

Work:

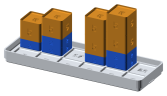
Answer: _____

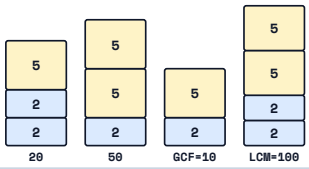
Section 1.5: Greatest Common Factor (GCF) & Least Common Multiple (LCM)

Big Ideas Method: Venn Diagram of Prime Factors — Shared Overlap for GCF & Tallest Prime Powers for LCM Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
When you stand two Prime Factor Towers next to each other on the 12-socket lid, you see GCF and LCM at the same time: GCF is the shared blocks both towers have in common (overlap), and LCM is the tallest tower needed to cover both (union)! Verify in 1 click with [GCF] and [f] [LCM].

BLOCK KIT & STEP-BY-STEP GUIDE (PRIME-01: PRIME FACTOR STUDIO)





HOW TO BUILD IT WITH PRIME-01 (PRIME FACTOR STUDIO):

Step 1: Open Kit PRIME-01 (Prime Factor Studio) and build Tower 20 ([2][2][5]) and Tower 50 ([2][5][5]) side by side.

Step 2: For GCF, pull out the prime blocks shared by BOTH towers: one [2] and one [5] -> [2][5] = 10.

Step 3: For LCM, build the tallest union covering both towers: two [2]s and two [5]s -> [2][2][5][5] = 100.

WORKED EXAMPLE — EXAMPLE 1: SIMULTANEOUS GCF & LCM OF 20 AND 50

Problem: Find both the GCF and LCM of 20 and 50 using Prime Towers and the SC-6.

Big Ideas Textbook Method (Venn Diagram of Prime Factors — Shared Overlap for GCF & Tallest Prim..

1. Write the prime factorization of each number: $20 = 2^2 \times 5$ and $50 = 2 \times 5^2$.
2. GCF (Shared Factors): Multiply the common prime factors with lesser exponents: $2^1 \times 5^1 = 10$.
3. LCM (All Factors): Multiply all prime factors using the greatest exponent of each: $2^2 \times 5^2 = 4 \times 25 = 100$.

Block Connection (PRIME-01): Tower 20 is [2][2][5] and Tower 50 is [2][5][5]. Both share [2][5] = 10 (GCF). Taking the max of each color gives [2][2][5][5] = 100 (LCM)!

SC-6 Keystrokes: `20` `ENTER` `50` `GCF` `20` `ENTER` `50` `f` `LCM`

Answer: GCF = 10, LCM = 100 *Check: GCF × LCM = 10 × 100 = 1,000, matching 20..*

SECTION 1.5 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Venn Diagram of Prime Factors — Shared Overlap for GCF & Talles..

1 Big Ideas Method: Prime Factor GCF & LCM
Find the GCF and LCM of 12 and 18 using Prime Towers and [GCF] / [f] [LCM].

Work:

Answer: _____

2 Big Ideas Method: Prime Factor GCF & LCM
Find the GCF and LCM of 8 and 12.

Work:

Answer: _____

3 Big Ideas Method: Prime Factor GCF & LCM
Find the GCF and LCM of 15 and 25.

Work:

Answer: _____

4 Big Ideas Method: Prime Factor GCF & LCM
Find the GCF and LCM of 14 and 35.

Work:

Answer: _____

5 Big Ideas Method: Distributive GCF Factoring
Find the GCF and LCM of 16 and 24.

Work:

Answer: _____

6 Big Ideas Method: Distributive GCF Factoring
Find the GCF and LCM of 28 and 42.

Work:

Answer: _____

7 Big Ideas Method: Distributive GCF Factoring
Two prime numbers are 5 and 7. What is their GCF and what is their LCM?

Work:

Answer: _____

8 Big Ideas Method: Distributive GCF Factoring
Verify that GCF(15, 20) × LCM(15, 20) equals 15 × 20.

Work:

Answer: _____

9 Application & 2-Way Check: GCF vs. LCM Word Problem
Hot dogs come in packs of 12 and buns in packs of 15. What is the smallest number of each to buy with 0 leftovers?

Work:

Answer: _____

10 Application & 2-Way Check: GCF vs. LCM Word Problem
You have 36 red tiles and 48 blue tiles to pack into identical kit bags with none left over. What is the max number of bags?

Work:

Answer: _____

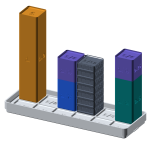
Section 2.1: Equivalent Fractions, Mixed Numbers & Adding Towers

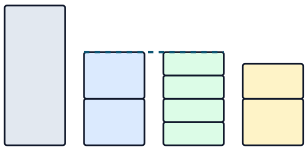
Big Ideas Method: Equivalent Fractions via GCF Simplification & Adding/Subtracting with Least Common Denominator.. **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

In the Fraction Studio (FRAC-01), 1 Whole is exactly 84.0 mm tall. Whenever two fraction towers stand at the exact same millimeter height, they are equivalent! Use [.] to enter fractions (e.g., [.] 4 [.] 6), [g] [SIMP] to simplify step-by-step, [f] [d/c] for improper fractions, and [FDISP] to toggle decimal view!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: 84MM FRACTION STUDIO)





HOW TO BUILD IT WITH FRAC-01 (84MM FRACTION STUDIO):

Step 1: Open Kit FRAC-01 (84mm Fraction Studio) and place the 84mm Whole (1) datum block on the comparator lid.

Step 2: Stack four 1/6 blocks (4 × 14mm = 56mm) next to two 1/3 blocks (2 × 28mm = 56mm) to see 12/18 = 4/6 = 2/3.

Step 3: To add 1/3 + 1/4, stack one 1/3 block (28mm) and one 1/4 block (21mm) = 49mm, matching seven 1/12 blocks (7/12).

WORKED EXAMPLE — EXAMPLE 1: STEP-BY-STEP SIMPLIFICATION WITH [G] [SIMP]

Problem: Simplify 12/18 step-by-step and compare tower heights.

Big Ideas Textbook Method (Equivalent Fractions via GCF Simplification & Adding/Subtracting with Le..

1. Divide numerator and denominator of 12/18 by common factors: (12 ÷ 2)/(18 ÷ 2) = 6/9, then (6 ÷ 3)/(9 ÷ 3) = 2/3.
2. To add unlike fractions (1/3 + 1/4), find the LCD of 3 and 4, which is 12.
3. Rewrite as equivalent twelfths and add numerators: 4/12 + 3/12 = 7/12.

Block Connection (FRAC-01): Four 1/6 blocks (4 × 14mm = 56mm) stand at the exact same height as two 1/3 blocks (2 × 28mm = 56mm).

SC-6 Keystrokes: [.] [12] [.] [18] [ENTER] [g] [SIMP] [g] [SIMP]

Answer: 12/18 -> 6/9 -> 2/3

Each press of [g] [SIMP] ([g][a b/c]) divides out one c..

SECTION 2.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Equivalent Fractions via GCF Simplification & Adding/Subtractin..

1 Big Ideas Method: Simplest Form / Mixed-Improper

Simplify 8/12 using [g] [SIMP]. What intermediate fraction appears first?

Work:

Answer: _____

2 Big Ideas Method: Simplest Form / Mixed-Improper

Simplify 24/36 completely and check its decimal with [FDISP].

Work:

Answer: _____

3 Big Ideas Method: Simplest Form / Mixed-Improper

Convert 2 3/5 into an improper fraction ([f] [d/c]) and a decimal ([FDISP]).

Work:

Answer: _____

4 Big Ideas Method: Simplest Form / Mixed-Improper

Convert 19/8 into a mixed number ([a b/c]) and a decimal ([FDISP]).

Work:

Answer: _____

5 Big Ideas Method: LCD Addition & Subtraction

Add 1/2 + 1/3 using a common denominator (sixths) and check with 84mm Fraction Blocks.

Work:

Answer: _____

6 Big Ideas Method: LCD Addition & Subtraction

Add 3/4 + 5/6 using the LCD (12). Write both the mixed number and improper fraction.

Work:

Answer: _____

7 Big Ideas Method: LCD Addition & Subtraction

Subtract 7/8 - 1/4 and verify with [f] [LASTX] [+].

Work:

Answer: _____

8 Big Ideas Method: LCD Addition & Subtraction

Subtract mixed numbers: 4 1/3 - 1 5/6.

Work:

Answer: _____

9 Application & 2-Way Check: Mixed Number Regrouping

Build 1/3 + 1/4 with 84mm towers (28mm + 21mm) AND solve using the LCD (twelfths).

Work:

Answer: _____

10 Application & 2-Way Check: Mixed Number Regrouping

A trail is 3 3/4 miles long. You hiked 1 7/8 miles. How much trail is left? Solve 2 ways!

Work:

Answer: _____

Section 2.2: Multiplying Fractions & Mixed Numbers

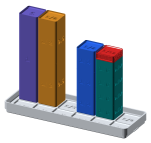
Big Ideas Method: Area/Tower Model for Fraction × Fraction & Cross-Canceling Common Factors

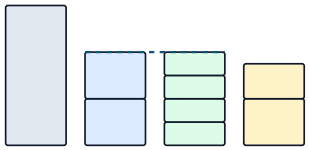
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Why does multiplying two proper fractions make a SMALLER answer? Because '×' means 'OF'! Taking 1/2 OF a 3/4 tower (63 mm) gives half its height (31.5 mm = 3/8). For mixed numbers, rewrite as improper fractions first (or use [f] [d/c]) and simplify common factors before multiplying!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: 84MM FRACTION STUDIO)





HOW TO BUILD IT WITH FRAC-01 (84MM FRACTION STUDIO):

Step 1: Open Kit FRAC-01 (84mm Fraction Studio) and stand three 1/4 blocks to build a 3/4 tower (63.0 mm tall).

Step 2: To find 1/2 OF 3/4, measure half the height of the 63.0 mm tower: 63.0 mm ÷ 2 = 31.5 mm.

Step 3: Place three 1/8 blocks (3 × 10.5 mm = 31.5 mm) alongside to prove 1/2 × 3/4 = 3/8.

WORKED EXAMPLE — EXAMPLE 1: FRACTION OF A FRACTION (1/2 × 3/4)

Problem: Find 1/2 × 3/4 using 84mm Fraction Towers and the SC-6.

Big Ideas Textbook Method (Area/Tower Model for Fraction × Fraction & Cross-Canceling Common Fa..

1. Multiply the numerators and multiply the denominators: (1 × 3) / (2 × 4) = 3/8.
2. For mixed numbers (1 1/2 × 2 2/3), convert to improper fractions first: 3/2 × 8/3.
3. Divide out common factors on the diagonal (cancel the 3s and divide 8 by 2) to get 4.

Block Connection (FRAC-01): Stand three 1/4 blocks (63 mm). Half of 63 mm is 31.5 mm, which matches three 1/8 blocks (3 × 10.5 mm = 3/8)!

SC-6 Keystrokes: [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [×]

Answer: 3/8

Check with [f] [LASTX] [÷] -> restores 1/2 immediately!

SECTION 2.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Area/Tower Model for Fraction × Fraction & Cross-Canceling Comm..

1 Big Ideas Method: Numerator × Numerator / Cross-Cancel

Find 1/2 × 2/3. (Hold two 1/3 blocks and take half of them!)

Work:

Answer: _____

2 Big Ideas Method: Numerator × Numerator / Cross-Cancel

Find 2/3 × 3/4 and check with [f] [LASTX] [÷].

Work:

Answer: _____

3 Big Ideas Method: Numerator × Numerator / Cross-Cancel

Multiply 3/5 × 5/9.

Work:

Answer: _____

4 Big Ideas Method: Numerator × Numerator / Cross-Cancel

Multiply a whole number by a fraction: 12 × 3/4.

Work:

Answer: _____

5 Big Ideas Method: Improper Fraction Conversion

Multiply mixed numbers: 1 3/4 × 2 2/7.

Work:

Answer: _____

6 Big Ideas Method: Improper Fraction Conversion

Multiply 2 1/4 × 1 1/3.

Work:

Answer: _____

7 Big Ideas Method: Improper Fraction Conversion

Multiply 3 1/2 × 2 1/2. Write the answer as a mixed number and decimal.

Work:

Answer: _____

8 Big Ideas Method: Improper Fraction Conversion

Prediction Check: Without calculating, is 5 × 2/3 greater than 5 or less than 5? Now compute it!

Work:

Answer: _____

9 Application & 2-Way Check: Scaling & Multi-Step Product

A slime recipe uses 1 3/4 cups of glue per batch. How much glue is needed for 2 1/2 batches?

Work:

Answer: _____

10 Application & 2-Way Check: Scaling & Multi-Step Product

Triple Chain: Evaluate 4/5 × 5/6 × 3/4 both by cross-canceling common factors AND on your SC-6!

Work:

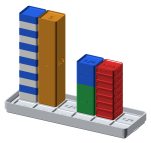
Answer: _____

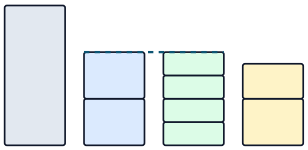
Section 2.3: Dividing Fractions & Mixed Numbers — Why Reciprocal Flip Works

Big Ideas Method: Measurement Division ('How Many Fit?') & Multiplying by the Reciprocal (Keep-Change-Flip) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Kids memorize 'Keep-Change-Flip' and then forget which fraction to flip! Physically, 3/4 ÷ 1/8 asks how many 1/8 blocks fit alongside a 3/4 tower (6 fit!). In Big Ideas Math, dividing by a fraction equals multiplying by its reciprocal—and [f] [1/x] flips ONLY the divisor into its reciprocal before you press [×]!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: 84MM FRACTION STUDIO)





HOW TO BUILD IT WITH FRAC-01 (84MM FRACTION STUDIO):

Step 1: Open Kit FRAC-01 (84mm Fraction Studio) and build a 3/4 dividend tower (three 21mm blocks = 63.0 mm).

Step 2: Stack 1/8 divisor blocks (10.5 mm each) right beside the 3/4 tower until the heights match.

Step 3: Count the 1/8 blocks: every 1/4 block matches two 1/8 blocks, so 3 × 2 = 6 blocks fit (3/4 ÷ 1/8 = 6).

WORKED EXAMPLE — EXAMPLE 1: PHYSICAL TOWER FIT VS. [F] [1/X] RECIPROCAL FLIP

Problem: Divide 3/4 ÷ 1/8 by comparing tower heights and by multiplying by the reciprocal with [f] [1/x].

Big Ideas Textbook Method (Measurement Division ('How Many Fit?') & Multiplying by the Reciprocal (..

1. Write the division problem: 3/4 ÷ 1/8.
2. Find the reciprocal of the divisor (flip 1/8 to 8/1) and rewrite division as multiplication: 3/4 × 8/1.
3. Simplify common factors (8 ÷ 4 = 2) and multiply: 3 × 2 = 6.

Block Connection (FRAC-01): One 1/4 block (21mm) holds two 1/8 blocks (10.5mm). So three 1/4 blocks hold 3 × 2 = 6 eighths!

SC-6 Keystrokes: `.` `3` `.` `4` `ENTER` `.` `1` `.` `8` `f` `1/x` `×`

Answer: 6

[f] [1/x] ([f][FDISP]) flips 1/8 into its reciprocal 8/1, the..

SECTION 2.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Measurement Division ('How Many Fit?') & Multiplying by the Rec..

1 Big Ideas Method: Multiply by Reciprocal
How many 1/4 blocks fit in a 1/2 tower? Compute 1/2 ÷ 1/4.

Work:

Answer: _____

2 Big Ideas Method: Multiply by Reciprocal
How many 1/6 blocks (14mm) fit in a 2/3 tower (56mm)? Compute 2/3 ÷ 1/6.

Work:

Answer: _____

3 Big Ideas Method: Multiply by Reciprocal
Divide 5/6 ÷ 5/12 by multiplying by the reciprocal ([f] [1/x] [×]).

Work:

Answer: _____

4 Big Ideas Method: Multiply by Reciprocal
Divide a whole number by a fraction: 6 ÷ 3/4.

Work:

Answer: _____

5 Big Ideas Method: Mixed-Number Reciprocal Division
Divide 3/4 ÷ 2/5. Write the answer as a mixed number.

Work:

Answer: _____

6 Big Ideas Method: Mixed-Number Reciprocal Division
Divide a fraction by a whole number: 4/5 ÷ 2.

Work:

Answer: _____

7 Big Ideas Method: Mixed-Number Reciprocal Division
Divide mixed numbers: 4 1/2 ÷ 3/4.

Work:

Answer: _____

8 Big Ideas Method: Mixed-Number Reciprocal Division
Divide 5 1/4 ÷ 1 3/4 and prove with [f] [LASTX] [×].

Work:

Answer: _____

9 Application & 2-Way Check: Equal-Size Measurement Cuts
A board is 6 3/4 ft long. You cut it into 3/4-ft shelves. How many shelves can you cut? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Equal-Size Measurement Cuts
A 3 3/4-cup pitcher of smoothie is poured into 5/8-cup glasses. How many glasses does it fill?

Work:

Answer: _____

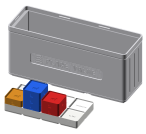
Section 2.4: Adding, Subtracting & Multiplying Decimals

Big Ideas Method: Vertical Decimal Point Alignment (+/-) & Counting Decimal Places / Estimation (×) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Paper decimal errors happen when place-value columns drift or decimal places are miscounted in multiplication. Use the Place Value Studio (PLACE-01) and the [f] [IP] (Integer Part) and [g] [FP] (Fractional Part) keys to align Tenths and Hundredths and estimate whole-number products first!

BLOCK KIT & STEP-BY-STEP GUIDE (PLACE-01: PLACE VALUE STUDIO)



Input: 14.85

[f] [IP]	[g] [FP]
14	0.85
Whole	Decimal

HOW TO BUILD IT WITH PLACE-01 (PLACE VALUE STUDIO):

Step 1: Open Kit PLACE-01 (Place Value Studio) and separate 14.85 into Whole Tens/Ones (14) and Tenths/Hundredths (0.85).

Step 2: To add 14.60 + 3.85, place 6 tenths + 8 tenths (regroup 10 tenths -> 1 one + 4 tenths) and 5 hundredths.

Step 3: Combine the whole ones (14 + 3 + 1 = 18) with the decimal part (0.45) to read 18.45.

WORKED EXAMPLE — EXAMPLE 1: ALIGNING DECIMAL PLACE VALUES ([F] [IP] & [G] [FP])

Problem: Split 14.85 into its whole and decimal parts, then compute 14.60 + 3.85.

Big Ideas Textbook Method (Vertical Decimal Point Alignment (+/-) & Counting Decimal Places / Estima..

1. Line up the decimal points vertically and annex a trailing zero so both numbers have hundredths: 14.60 + 3.85.
2. Add column by column from right to left (hundredths, tenths, ones, tens): 14.60 + 3.85 = 18.45.
3. Check by subtracting 18.45 - 3.85 = 14.60.

Block Connection (PLACE-01): On the Place Value tray, 14 sits in the Tens/Ones bays ([f] [IP] -> 14) and .85 sits in Tenths/Hundredths ([g] [FP] -> 0.85).

SC-6 Keystrokes: `14.6` `ENTER` `3.85` `+` `f` `LASTX` `-`

Answer: 18.45 (Checked: 18.45 - 3.85. *[f] [IP] ([f][PV]) shows whole part; [g] [FP] ([g][PV]) sh..*

SECTION 2.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Vertical Decimal Point Alignment (+/-) & Counting Decimal Place..

1 Big Ideas Method: Vertical Decimal Alignment

Enter 27.64. Use [f] [IP] and [g] [FP] to write its Whole part and Decimal part.

Work:

Answer: _____

2 Big Ideas Method: Vertical Decimal Alignment

Line up the decimal points to add 12.8 + 4.75, and check with [f] [LASTX] [-].

Work:

Answer: _____

3 Big Ideas Method: Vertical Decimal Alignment

Annex zeros to subtract 20.00 - 8.65.

Work:

Answer: _____

4 Big Ideas Method: Vertical Decimal Alignment

Multiply 4.5 × 0.6.

Work:

Answer: _____

5 Big Ideas Method: Decimal Place Count & Estimate

Evaluate 25.4 - 9.86 + 3.2.

Work:

Answer: _____

6 Big Ideas Method: Decimal Place Count & Estimate

Multiply 3.6 × 2.25 and check the whole-number ballpark.

Work:

Answer: _____

7 Big Ideas Method: Decimal Place Count & Estimate

Multiply 0.15 × 0.4 and tap [FDISP] to see its simplified fraction.

Work:

Answer: _____

8 Big Ideas Method: Decimal Place Count & Estimate

Find the area of a rectangular desk top measuring 1.8 m by 0.75 m.

Work:

Answer: _____

9 Application & 2-Way Check: Multi-Step Money & Area

You pay with a $20 bill for a $8.75 burger and $4.60 smoothie. How much change do you get?

Work:

Answer: _____

10 Application & 2-Way Check: Multi-Step Money & Area

Multiply 4.75 × 1.2 both as decimals AND as fractions (4 3/4 × 1 1/5) using [FDISP]!

Work:

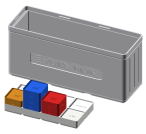
Answer: _____

Section 2.5: Dividing Decimals & The 3-Way Fraction-Decimal-Remainder Translator

Big Ideas Method: Multiplying Dividend & Divisor by a Power of 10 to Make a Whole-Number Divisor Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Why are 15 ÷ 4 = 3 R 3, 3 3/4, and 3.75 all the same answer? Because 3 leftover pieces divided by 4 is 3/4 = 0.75! On the SC-6, tapping [FDISP] after any division cycles through Decimal, Mixed Fraction, Improper Fraction, and Quotient+Remainder with one thumb!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: FRACTION & PLACE VALUE STUDIO)



Input: 15 ÷ 4

[f] [IP]	[g] [FP]
3 R 3	3.75 = 3 3/4
Whole	Decimal

HOW TO BUILD IT WITH FRAC-01 (FRACTION & PLACE VALUE STUDIO):

Step 1: Open Kit FRAC-01 / PLACE-01 and model 4.5 ÷ 0.75 as measuring how many 0.75 (3/4) towers fit into 4.5 (4 1/2).

Step 2: Shift both decimals 2 places right (× 100) by trading 4.50 wholes into 450 hundredths and 0.75 into 75 hundredths.

Step 3: Group 450 hundredths into sets of 75 hundredths: 6 complete groups fit (450 ÷ 75 = 6).

WORKED EXAMPLE — EXAMPLE 1: THE 4-OUTFIT [FDISP] TRANSLATOR (23 ÷ 4)

Problem: Write 23 ÷ 4 as a decimal, mixed number, improper fraction, and quotient with remainder.

Big Ideas Textbook Method (Multiplying Dividend & Divisor by a Power of 10 to Make a Whole-Number..

1. Multiply both the divisor (0.75) and dividend (4.50) by 100 so the divisor is a whole number: 450 ÷ 75.
2. Divide 450 ÷ 75 = 6.
3. Check by multiplying Quotient × Original Divisor: 6 × 0.75 = 4.5.

Block Connection (FRAC-01): 23 quarter-blocks make 5 whole stacks of 4 (20) plus 3 fourths (3/4 = 0.75).

SC-6 Keystrokes: 23 ENTER 4 ÷ FDISP f d/c

Answer: 5.75 = 5 3/4 = 23/4 = 5 R 3 *Tap [FDISP] repeatedly to see every textbook format..*

SECTION 2.5 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Multiplying Dividend & Divisor by a Power of 10 to Make a Whole..

1 Big Ideas Method: Power-of-10 Divisor Shift
Write 39 ÷ 8 as (a) Quotient + Remainder, (b) Decimal, and (c) Mixed Number.

Work:

Answer: _____

2 Big Ideas Method: Power-of-10 Divisor Shift
Divide a decimal by a whole number: 18.4 ÷ 8.

Work:

Answer: _____

3 Big Ideas Method: Power-of-10 Divisor Shift
How many quarters ($0.25) are in $11.25? Compute 11.25 ÷ 0.25.

Work:

Answer: _____

4 Big Ideas Method: Power-of-10 Divisor Shift
Divide 7.2 ÷ 0.9 and check with [f] [LASTX] [×].

Work:

Answer: _____

5 Big Ideas Method: Decimal-Fraction Equivalence
Divide 15.6 ÷ 0.12.

Work:

Answer: _____

6 Big Ideas Method: Decimal-Fraction Equivalence
Divide 9.45 ÷ 1.5 and tap [FDISP] to write its mixed fraction form.

Work:

Answer: _____

7 Big Ideas Method: Decimal-Fraction Equivalence
Divide 0.84 ÷ 0.07.

Work:

Answer: _____

8 Big Ideas Method: Decimal-Fraction Equivalence
Spot the Glitch: Leo says 4.8 ÷ 0.2 = 0.24. Use [f] [LASTX] or estimation to fix it!

Work:

Answer: _____

9 Application & 2-Way Check: Unit Price & Packaging Division
A runner completes a 4.5-mile race in 0.75-mile laps. How many laps is the race? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Unit Price & Packaging Division
A 6-pack of sports drinks costs $7.50. What is the cost of 1 bottle, and how many bottles can $20 buy?

Work:

Answer: _____

Section 3.1: Ratios & Tape Diagrams — Part-to-Part vs. Part-to-Whole

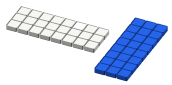
Big Ideas Method: Part-to-Part vs. Part-to-Whole Tape Diagrams & Equal-Share Box Value

Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Instead of drawing uneven paper boxes for a Tape Diagram, snap Blue and Gold Counting Tiles onto your desk! Each tile is 1 equal unit box. For a 3:2 ratio, 3 Blue + 2 Gold = 5 total tiles (Part-to-Whole = 3/5). Use [g] [a:b] ([g][GCF]) to reduce any ratio to simplest form!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)



Part A (3 tiles):

4 4 4

Part B (2 tiles):

4 4

Ratio 3:2 (5 boxes total)

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and place 3 blue tiles (Row A) above 2 gold tiles (Row B) for 3:2.

Step 2: To share 20 items in a 3:2 ratio, count all 3 + 2 = 5 boxes on the tape diagram.

Step 3: Divide 20 ÷ 5 = 4 per box; label each tile '4' to read Part A = 3 × 4 = 12 and Part B = 2 × 4 = 8.

WORKED EXAMPLE — EXAMPLE 1: SIMPLIFYING RATIOS WITH [G] [A:B]

Problem: A gaming clan has 18 builders and 12 speedrunners. Simplify the Part-to-Part and Part-to-Whole ratios.

Big Ideas Textbook Method (Part-to-Part vs. Part-to-Whole Tape Diagrams & Equal-Share Box Value):

1. Draw a Tape Diagram with 3 boxes for Part A and 2 boxes for Part B (Total = 3 + 2 = 5 equal boxes).
2. Divide the total quantity by total boxes to find the value of 1 box: 20 ÷ 5 = 4 per box.
3. Multiply by the number of boxes in each part: Part A = 3 × 4 = 12, Part B = 2 × 4 = 8.

Block Connection (COUNT-01): Group 18 and 12 by their GCF (6): you get 3 builders for every 2 speedrunners (and 3 + 2 = 5 players per group).

SC-6 Keystrokes: 18 ENTER 12 g a:b

Answer: Part-to-Part = 3:2; Part-to-.. *[g] [a:b] divides both Y and X by their GCF in 1 click!*

SECTION 3.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Part-to-Part vs. Part-to-Whole Tape Diagrams & Equal-Share Box..

1 Big Ideas Method: Simplified Ratio (a : b)

Simplify the ratio 12 : 18 using [g] [a:b].

Work:

Answer: _____

2 Big Ideas Method: Simplified Ratio (a : b)

Simplify the ratio 24 : 16 using [g] [a:b].

Work:

Answer: _____

3 Big Ideas Method: Simplified Ratio (a : b)

A bag has 4 red marbles and 6 blue marbles. Find the simplified ratio of red to TOTAL marbles.

Work:

Answer: _____

4 Big Ideas Method: Simplified Ratio (a : b)

In a 3:4 tape diagram, each tile is worth 6. Find the two parts and the total.

Work:

Answer: _____

5 Big Ideas Method: Tape Diagram Box Value

Green paint mixes blue and yellow in a 2:3 ratio. If you use 25 cups total, how many cups of each?

Work:

Answer: _____

6 Big Ideas Method: Tape Diagram Box Value

The ratio of cats to dogs at a shelter is 4:5. There are 36 animals total. How many are dogs?

Work:

Answer: _____

7 Big Ideas Method: Tape Diagram Box Value

The ratio of boys to girls in a club is 3:5. If there are 15 boys, how many girls are there?

Work:

Answer: _____

8 Big Ideas Method: Tape Diagram Box Value

Difference Tape Model: The ratio of acoustic to electric guitars is 2:5. There are 18 MORE electric guitars. Find how many electric guitars!

Work:

Answer: _____

9 Application & 2-Way Check: Part-to-Whole Tape Split

Two robot teams split 84 gears in a 4:3 ratio. How many gears does each team get? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Part-to-Whole Tape Split

Ratio of gold to silver coins is 7:4. There are 21 more gold coins than silver. How many coins in all?

Work:

Answer: _____

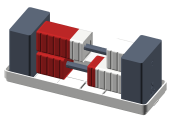
Section 3.2: Ratio Tables & Double Number Lines via [g] [REP]

Big Ideas Method: Equivalent Ratio Tables — Multiplying or Dividing Both Quantities by the Same Scale Factor Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

A Ratio Table and a Double Number Line are the exact same thing: two skip-counting ladders running side-by-side! Press 3 [g] [REP] and tap [+] to march across the top row (3, 6, 9, 12, 15), then 7 [g] [REP] and tap [+] to march across the bottom row (7, 14, 21, 28, 35)!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)



Part A (3 tiles):



Part B (5 tiles):



Ratio 3:5 (8 boxes total)

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and lay down Row 1 of a ratio table: 3 blue tiles and 4 gold tiles.

Step 2: Add identical rows of (3 blue, 4 gold) underneath to scale the ratio table by ×2 (6:8), ×3 (9:12), and ×6 (18:24).

Step 3: Notice that adding a new row adds +3 to Column A and +4 to Column B, keeping the ratio 3:4 constant!

WORKED EXAMPLE — EXAMPLE 1: GENERATING A RATIO TABLE WITH [G] [REP]

Problem: A drone flies 8 meters every 3 seconds. Build a 5-step table to find how far it flies in 15 seconds.

Big Ideas Textbook Method (Equivalent Ratio Tables — Multiplying or Dividing Both Quantities by the..

1. Set up a Ratio Table with columns for Quantity A (3) and Quantity B (4).
2. Find the scale factor from 3 to 18: 18 ÷ 3 = 6 (multiply by 6).
3. Multiply the second quantity by the same scale factor: 4 × 6 = 24, so 3:4 is equivalent to 18:24.

Block Connection (COUNT-01): Seconds ladder: 3 [g] [REP] [+] [+] [+] [+] -> 3, 6, 9, 12, 15 (5 jumps). Meters ladder: 8 [g] [REP] [+] [+] [+] [+] -> 8, 16, 24, 32, 40!

SC-6 Keystrokes: 8 g REP + + + +

Answer: 40 meters in 15 seconds

[g] [REP] ([g][x↔y]) locks the step size so each [+] ad..

SECTION 3.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Equivalent Ratio Tables — Multiplying or Dividing Both Quantiti..

1 Big Ideas Method: Scale Factor Multiplication

Use 4 [g] [REP] and 9 [g] [REP] to write the first 4 equivalent ratios of 4 : 9.

Work:

Answer: _____

2 Big Ideas Method: Scale Factor Multiplication

In a ratio table where A : B = 4 : 9, find B when A = 28.

Work:

Answer: _____

3 Big Ideas Method: Scale Factor Multiplication

In a ratio table where X : Y = 5 : 12, find X when Y = 60.

Work:

Answer: _____

4 Big Ideas Method: Scale Factor Multiplication

A printer prints 15 pages every 2 minutes. How many pages in 8 minutes?

Work:

Answer: _____

5 Big Ideas Method: Ratio Table Missing Entry

Hot cocoa uses 3 tbsp cocoa for every 8 oz milk. How much milk for 18 tbsp cocoa?

Work:

Answer: _____

6 Big Ideas Method: Ratio Table Missing Entry

8 markers cost $12. Scale down to 2 markers, then scale up to find the cost of 10 markers.

Work:

Answer: _____

7 Big Ideas Method: Ratio Table Missing Entry

A heart beats 21 times in 15 seconds. How many beats in 60 seconds (1 minute)?

Work:

Answer: _____

8 Big Ideas Method: Ratio Table Missing Entry

Spot the Glitch: For 3:5, Sam added 4 to both sides and wrote 7:9. Test with [g] [a:b]—are they equal?

Work:

Answer: _____

9 Application & 2-Way Check: Two-Step Ratio Table Jump

9 smoothies take 12 cups of berries. How many cups of berries for 15 smoothies? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Two-Step Ratio Table Jump

An e-bike travels 14 miles on 35% battery. How far can it travel on 100% battery?

Work:

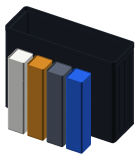
Answer: _____

Section 3.3: Unit Rates & Best-Buy Showdowns (Plus [x↔y] Swap Rescue)

Big Ideas Method: Unit Rate Method — Dividing Both Terms by the Second Term to Find Amount per 1 Unit Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
'Dollars PER Ounce' means Dollars ÷ Ounces (enter Dollars on Y, Ounces on X, press [÷]). What if you accidentally typed Ounces first and Dollars second? Don't clear! Just press [x↔y] ([f][R↓]) to swap Floor 1 and Floor 2, then press [÷]!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)



GROUP 1 TILES:
Bag A: $4.20 ÷ 12 = $0.35/oz

GROUP 2 TILES:
Bag B: $5.12 ÷ 16 = $0.32/oz

==> Bag B Wins!

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit PLACE-01 / COUNT-01 and model 180 miles across 3 equal hour-blocks.

Step 2: Divide the total miles equally among the 3 hour-blocks: 180 ÷ 3 = 60 miles on a single 1-hour block.

Step 3: Scale the 1-hour unit rate block by 5 blocks to find the distance in 5 hours: 60 × 5 = 300 miles.

WORKED EXAMPLE — EXAMPLE 1: THE [X↔Y] UPSIDE-DOWN RESCUE

Problem: Find the unit rate in words per minute for 315 words in 5 minutes (even if you typed 5 first!).

Big Ideas Textbook Method (Unit Rate Method — Dividing Both Terms by the Second Term to Find Am..

1. Write the rate as a fraction with the 'per' unit in the denominator: 180 miles / 3 hours.
2. Divide numerator and denominator by 3 to get a denominator of 1: 60 miles / 1 hour (60 mph).
3. Multiply the unit rate by 5 hours: 60 mph × 5 hr = 300 miles.

Block Connection (COUNT-01): If you typed 5 [ENTER] 315, tap [x↔y] so Y=315 and X=5, then press [÷]!

SC-6 Keystrokes: `5` `ENTER` `315` `x↔y` `÷`

Answer: 63 words per minute

[x↔y] swaps Floor 1 (X) and Floor 2 (Y) instantly!

SECTION 3.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Unit Rate Method — Dividing Both Terms by the Second Term to Fi..

1 Big Ideas Method: Unit Rate (Amount per 1)
Find the unit speed in miles per hour: 195 miles in 3 hours.

Work:

Answer: _____

2 Big Ideas Method: Unit Rate (Amount per 1)
You typed 4 [ENTER] 36 for $36 in 4 hours. Press [x↔y] [÷] to find the hourly pay!

Work:

Answer: _____

3 Big Ideas Method: Unit Rate (Amount per 1)
Find the unit price per pound: $7.80 for 6 pounds of apples.

Work:

Answer: _____

4 Big Ideas Method: Unit Rate (Amount per 1)
A 3D printer uses 90 grams of filament in 6 hours. How many grams per hour?

Work:

Answer: _____

5 Big Ideas Method: Unit Rate Scaling (Rate × Units)
Best Buy: Cereal A is $4.50 for 15 oz. Cereal B is $5.04 for 18 oz. Find both unit prices!

Work:

Answer: _____

6 Big Ideas Method: Unit Rate Scaling (Rate × Units)
Best Buy: 6-pack of pens for $5.10 vs. 10-pack of pens for $8.20.

Work:

Answer: _____

7 Big Ideas Method: Unit Rate Scaling (Rate × Units)
A train travels 210 miles in 3.5 hours. At this rate, how far does it travel in 5 hours?

Work:

Answer: _____

8 Big Ideas Method: Unit Rate Scaling (Rate × Units)
Pump A fills 45 gallons in 5 min. Pump B fills 64 gallons in 8 min. Which pump is faster?

Work:

Answer: _____

9 Application & 2-Way Check: Best-Buy Unit Price Comparison
An extruder pushes 45 mm of filament in 15 seconds. How much in 60 seconds (1 min)? Solve 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Best-Buy Unit Price Comparison
Car A drives 168 miles on 6 gallons. Car B drives 232 miles on 8 gallons. Which car gets more mpg?

Work:

Answer: _____

Section 3.4: Percents, Fractions & Decimals — The 3-Costume Switcher

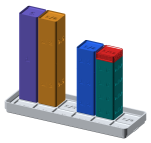
Big Ideas Method: Graphing Ratio Tables as Ordered Pairs (x, y) & Comparing Steepness (Unit Rate)

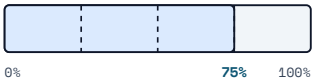
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

'Per-cent' literally means 'out of 100' (like 100 cents in $1.00). Instead of memorizing rules for moving decimal points left or right, enter any fraction or decimal on the SC-6 and press [f] [%] ([f][÷]) to see its percent, or [FDISP] to switch between fraction and decimal!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: 84MM FRACTION STUDIO)





HOW TO BUILD IT WITH FRAC-01 (84MM FRACTION STUDIO):

Step 1: Open Kit COUNT-01 (4x4 Gridfinity Baseplate) and treat the bottom edge as the x-axis and left edge as the y-axis.

Step 2: Snap tiles at (1, 3), (2, 6), and (3, 9) by walking right x columns and climbing up y rows.

Step 3: Lay a straightedge across the tiles—every ratio table forms a straight line through the origin (0, 0)!

WORKED EXAMPLE — EXAMPLE 1: FRACTION <-> DECIMAL <-> PERCENT IN 1 CLICK

Problem: Convert 7/20 into a decimal and a percent using [FDISP] and [f] [%].

Big Ideas Textbook Method (Graphing Ratio Tables as Ordered Pairs (x, y) & Comparing Steepness (U..

1. Build a Ratio Table of (x, y) pairs from the unit rate: (1, 3), (2, 6), (3, 9), (4, 12).
2. Plot each ordered pair (x, y) on the coordinate plane and connect them from (0, 0).
3. Compare two lines: at x = 1, the line with the greater y-value (greater unit rate) is steeper.

Block Connection (FRAC-01): Since 20 × 5 = 100, 7/20 is 35/100 = 0.35 = 35%. Verify with [FDISP] and [f] [%].

SC-6 Keystrokes: [.] [7] [.] [20] [ENTER] [FDISP] [100] [×]

Answer: 0.35 and 35%

To turn any percent p% into a fraction: key p [ENTER..

SECTION 3.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Graphing Ratio Tables as Ordered Pairs (x, y) & Comparing Steep..

1 Big Ideas Method: Ordered Pairs from Ratio Table

Write 3/5 as a decimal and as a percent.

Work:

Answer: _____

2 Big Ideas Method: Ordered Pairs from Ratio Table

Write 9/25 as a decimal and as a percent.

Work:

Answer: _____

3 Big Ideas Method: Ordered Pairs from Ratio Table

Write 45% as a decimal and as a simplified fraction ([FDISP]).

Work:

Answer: _____

4 Big Ideas Method: Ordered Pairs from Ratio Table

Write 0.125 as a percent and as a simplified fraction ([FDISP]).

Work:

Answer: _____

5 Big Ideas Method: Unit Rate from Point (1, r)

Find 30% of 80 using the [f] [%] key: 80 [ENTER] 30 [f] [%].

Work:

Answer: _____

6 Big Ideas Method: Unit Rate from Point (1, r)

Find 15% of 60.

Work:

Answer: _____

7 Big Ideas Method: Unit Rate from Point (1, r)

You answered 21 out of 25 questions right. What is your score as a percent?

Work:

Answer: _____

8 Big Ideas Method: Unit Rate from Point (1, r)

Convert 150% into a decimal and a mixed fraction.

Work:

Answer: _____

9 Application & 2-Way Check: Steepness / Speed Comparison

Order 5/8, 60%, and 0.61 from least to greatest by converting all 3 to decimals (and percents!).

Work:

Answer: _____

10 Application & 2-Way Check: Steepness / Speed Comparison

A phone battery is at 35% of its 4,000 mAh capacity. How many mAh remain? Solve 2 ways!

Work:

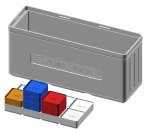
Answer: _____

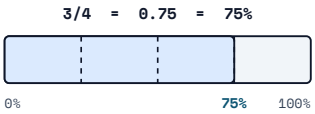
Section 3.5: Discounts, Percent Change ([g] [%CHG]) & Unit Conversions

Big Ideas Method: Unit Analysis (Multiplying by a Conversion Factor That Cancels Units) & Base-60 Time **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
On a normal calculator, finding a sale price takes two separate calculations because computing the discount erases the original price! On the SC-6, typing 60 [ENTER] 25 [f] [%] puts the $15 discount on Floor 1 (X) while keeping $60 on Floor 2 (Y)—so pressing [-] gives the $45 sale price in 1 click!

BLOCK KIT & STEP-BY-STEP GUIDE (PLACE-01: PLACE VALUE & FRACTION STUDIO)





HOW TO BUILD IT WITH PLACE-01 (PLACE VALUE & FRACTION STUDIO):

Step 1: Open Kit TIME-01 (Planetary Geared Clock) and turn the minute hand 1 full revolution (60 minutes).

Step 2: Watch the internal 12:1 planetary gear train advance the hour hand by exactly 1 hour mark (60 min = 1 hr).

Step 3: Dial 1 hr 45 min and advance 50 min past the 60-minute top mark to read 2 hr 35 min directly.

WORKED EXAMPLE — EXAMPLE 1: 2-KEY SALE PRICE ([F] [%] THEN [-])

Problem: A $60 video game is 25% off. Find the discount and the sale price.

Big Ideas Textbook Method (Unit Analysis (Multiplying by a Conversion Factor That Cancels Units) &..

1. Write the given measurement and multiply by the conversion factor that cancels the starting unit: 36 ft × (1 yd / 3 ft) = 12 yd.
2. For Base-60 time (1 hr 45 min + 50 min), add minutes: 45 + 50 = 95 min.
3. Regroup 60 minutes into 1 hour: 95 min = 1 hr 35 min -> Total = 2 hr 35 min.

Block Connection (PLACE-01): 25% is 1/4 of $60 = $15 off. The remaining 75% (3/4) is $45!

SC-6 Keystrokes: 60 | ENTER | 25 | f | % | -

Answer: Discount = $15; Sale Price =.. *For tax or tip, press [+] instead of [-] at the end!*

SECTION 3.5 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Unit Analysis (Multiplying by a Conversion Factor That Cancels..

1 Big Ideas Method: Unit Conversion Factor
An $80 pair of shoes is 30% off. Find the discount (X) and sale price ([-]).

Work:

Answer: _____

2 Big Ideas Method: Unit Conversion Factor
A $40 restaurant meal has a 15% tip. Find the tip and total bill ([+]).

Work:

Answer: _____

3 Big Ideas Method: Unit Conversion Factor
Find the percent change from 50 to 65 using [g] [%CHG].

Work:

Answer: _____

4 Big Ideas Method: Unit Conversion Factor
Find the percent change from 80 down to 60 using [g] [%CHG].

Work:

Answer: _____

5 Big Ideas Method: Base-60 Time Regrouping
Convert 10 inches to centimeters using [f] [->cm] (2.54 cm = 1 in).

Work:

Answer: _____

6 Big Ideas Method: Base-60 Time Regrouping
Convert 50.8 centimeters back to inches using [g] [->in] ([g][1]).

Work:

Answer: _____

7 Big Ideas Method: Base-60 Time Regrouping
Convert 11 pounds to kilograms using [f] [->kg] ([f][3]).

Work:

Answer: _____

8 Big Ideas Method: Base-60 Time Regrouping
Reverse Percent: 18 is 25% of what number? (Divide 18 by 0.25!)

Work:

Answer: _____

9 Application & 2-Way Check: Rate & Unit Conversion
A $120 skateboard is 20% off. Find the sale price both by subtracting [f][%] AND by multiplying by 80%!

Work:

Answer: _____

10 Application & 2-Way Check: Rate & Unit Conversion
A 32-inch deck weighs 5 lbs. Convert length to cm ([f][->cm]) and weight to kg ([f][->kg]). Verify with ×2.54!

Work:

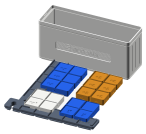
Answer: _____

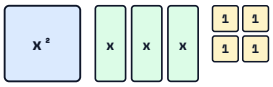
Section 4.1: Evaluating Algebraic Expressions — Variable Boxes on the Stack

Big Ideas Method: 10×10 Hundredths Grid ('Per 100'), Decimal-Point Shift & Fraction Benchmarks Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
A letter like x is not scary—it is just an empty green box (ALGB-01) waiting for you to drop a number inside! When you see 4x + 7 and x = 5, replace each x-rod with 5 units (4 × 5 = 20) and add 7 -> 27. On the SC-6, evaluating an expression is just Section 1.3's elevator math!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & TILES)





Sort by Shape:
x²-Flats | x-Rods | 1-Cubes

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & TILES):
Step 1: Open Kit FRAC-01 / PLACE-01 and model 100% as the full 100-square hundredths grid (or full 84mm Whole tower).
Step 2: Shade 35 of the 100 squares to represent 35% = 35/100 = 0.35.
Step 3: Group the 35 shaded squares and 100 total squares by 5s to see the simplified fraction 7/20.

WORKED EXAMPLE — EXAMPLE 1: SINGLE-VARIABLE PLUG-IN (4X + 7 WHEN X = 5)

Problem: Evaluate 4x + 7 when x = 5 using Algebra Tiles and the SC-6.

Big Ideas Textbook Method (10×10 Hundredths Grid ('Per 100'), Decimal-Point Shift & Fraction Benchm..
1. Percent to Decimal: Divide by 100 (move decimal 2 places left): 35% = 35 ÷ 100 = 0.35.
2. Percent to Fraction: Write over 100 and simplify by GCF(35, 100) = 5: 35/100 = 7/20.
3. Fraction to Percent: Convert 3/8 = 0.375, then multiply by 100 -> 37.5%.

Block Connection (ALGB-01): Lay out 4 green x-rods and 7 unit cubes. Replace each x-rod with 5 units (4 × 5 = 20), then add 7 units -> 27!

SC-6 Keystrokes: 4 | ENTER | 5 | × | 7 | +

Answer: 27 *Write Snapshot Y: 20 | X: 7 before pressing [+]!*

SECTION 4.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: 10×10 Hundredths Grid ('Per 100'), Decimal-Point Shift & Fracti..

1 Big Ideas Method: Per-100 Grid / Decimal Shift
Evaluate 6x - 9 when x = 7.
Work:
Answer: _____

2 Big Ideas Method: Per-100 Grid / Decimal Shift
Evaluate 5n + 14 when n = 8.
Work:
Answer: _____

3 Big Ideas Method: Per-100 Grid / Decimal Shift
Evaluate 40 - 3m when m = 9. (Park 40 on Y first!)
Work:
Answer: _____

4 Big Ideas Method: Per-100 Grid / Decimal Shift
Evaluate $x^2 + 11$ when x = 6.
Work:
Answer: _____

5 Big Ideas Method: Fraction <-> Percent Conversion
Evaluate $2y^2 - 5$ when y = 4. (Square 4 before multiplying by 2!)
Work:
Answer: _____

6 Big Ideas Method: Fraction <-> Percent Conversion
Evaluate 7a + 3b when a = 4 and b = 9.
Work:
Answer: _____

7 Big Ideas Method: Fraction <-> Percent Conversion
Evaluate (x + y) ÷ 4 when x = 19 and y = 13.
Work:
Answer: _____

8 Big Ideas Method: Fraction <-> Percent Conversion
Evaluate 5k - 2.4 when k = 3.5.
Work:
Answer: _____

9 Application & 2-Way Check: 3-Way Equivalent Form
Skate park rental costs $8 flat plus $4.50 per hour (8 + 4.50h). Find the cost for 4 hours 2 ways!
Work:
Answer: _____

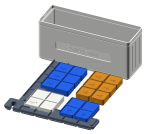
10 Application & 2-Way Check: 3-Way Equivalent Form
Surface area of a cube is $6s^2$. Find the surface area when edge s = 5 cm.
Work:
Answer: _____

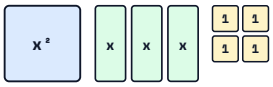
Section 4.2: Combining Like Terms — Why x² Flats, x Rods & 1 Cubes Never Mix

Big Ideas Method: Ordering Fractions, Decimals & Percents by Converting to a Common Form (Decimals or Percents) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Why can't you add 3x + 2 and write 5x? Hold a green x-rod (42 mm long) next to a small yellow 1-cube (14 mm)—they are completely different shapes! 'Combining Like Terms' simply means sorting your tiles into separate piles by shape (x²-flats, x-rods, and 1-cubes) and counting each pile!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & TILES)





Sort by Shape:
x²-Flats | x-Rods | 1-Cubes

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & TILES):

Step 1: Open Kit FRAC-01 (84mm Fraction Studio) and stand the 3/5 tower alongside the hundredths/decimal scale.

Step 2: Convert each number into the same 'costume' (decimals or percents): 3/5 = 0.60 (60%), 0.58 (58%), and 62% (0.62).

Step 3: Line them up on the number line from least to greatest: 0.58 < 3/5 < 62%.

WORKED EXAMPLE — EXAMPLE 1: SORTING X-RODS AND UNIT CUBES (4X + 5 + 2X + 3)

Problem: Simplify 4x + 5 + 2x + 3 and prove it by testing x = 10 on your SC-6.

Big Ideas Textbook Method (Ordering Fractions, Decimals & Percents by Converting to a Common For..

1. Rewrite every number in the same form (decimals): 3/5 = 0.60, 0.58 = 0.58, and 62% = 0.62.
2. Compare the hundredths digits on a number line: 0.58 < 0.60 < 0.62.
3. Write the final order using the original numbers: 0.58, 3/5, 62%.

Block Connection (ALGB-01): Slide the x-rods together (4x + 2x = 6x) and slide the unit cubes together (5 + 3 = 8) -> 6x + 8.

SC-6 Keystrokes: `4` `ENTER` `2` `+` `5` `ENTER` `3` `+`

Answer: 6x + 8

Test x=10: Original 4(10)+5+2(10)+3 = 68; Simplified..

SECTION 4.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Ordering Fractions, Decimals & Percents by Converting to a Comm..

1 Big Ideas Method: Common Decimal/Percent Form
Sort and simplify: 7x + 4 + 2x + 9.

Work:

Answer: _____

2 Big Ideas Method: Common Decimal/Percent Form
Simplify: 8x + 11 - 3x - 4.

Work:

Answer: _____

3 Big Ideas Method: Common Decimal/Percent Form
Simplify: 9y - y + 6. (Remember: -y means -1y!)

Work:

Answer: _____

4 Big Ideas Method: Common Decimal/Percent Form
Simplify: 12a + 5 - 4a + 8.

Work:

Answer: _____

5 Big Ideas Method: Least-to-Greatest Ordering
Sort all 3 tile shapes: $4x^2 + 3x + 2x^2 + 5x + 7$.

Work:

Answer: _____

6 Big Ideas Method: Least-to-Greatest Ordering
Simplify: $6n^2 + 9n - 2n^2 - 4n + 10$.

Work:

Answer: _____

7 Big Ideas Method: Least-to-Greatest Ordering
Simplify with decimals: 3.5x + 4.2 + 1.8x - 1.5.

Work:

Answer: _____

8 Big Ideas Method: Least-to-Greatest Ordering
Spot the Glitch: Leo simplified 5x + 3 to 8x. Test x = 10 on your SC-6 to prove 5x + 3 != 8x!

Work:

Answer: _____

9 Application & 2-Way Check: Multi-Form Comparison
A rectangle has two sides of (2x + 3) and two sides of (x + 5). Write its simplified perimeter & test x=10!

Work:

Answer: _____

10 Application & 2-Way Check: Multi-Form Comparison
A triangle has sides (3x + 2), (2x + 7), and (4x - 1). Write the simplified perimeter & test x=5!

Work:

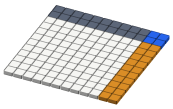
Answer: _____

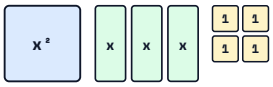
Section 4.3: The Distributive Property — One Split Rectangle, Two Ways

Big Ideas Method: Double Number Line, 10% Mental Benchmark & Percent Equation (Part = Percent × Whole) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Why does 3(x + 4) = 3x + 12? Build 3 identical rows of (1 green x-rod + 4 yellow cubes) on the Algebra Stage. Count by rows: 3(x + 4). Count by shape columns: 3 x-rods + 12 cubes = 3x + 12! Same tiles on your desk, two ways to name the area!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & TILES)





Sort by Shape:
x²-Flats | x-Rods | 1-Cubes

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & TILES):
Step 1: Open Kit TENF-01 / PLACE-01 and lay out a 10-part Percent Bar where the full 100% bar equals 80.
Step 2: Divide 80 across the 10 equal 10%-blocks: each 10% block is worth 8, and half a block (5%) is worth 4.
Step 3: Combine three 10% blocks (3 × 8 = 24) and one 5% half-block (+ 4) to get 35% of 80 = 28.

WORKED EXAMPLE — EXAMPLE 1: NUMERICAL SPLIT RECTANGLE 6(10 + 4)

Problem: Compute 6 × (10 + 4) both ways on your SC-6 to see why distribution works.

Big Ideas Textbook Method (Double Number Line, 10% Mental Benchmark & Percent Equation (Part =..
1. Benchmark Method: 10% of 80 = 8, so 30% = 3 × 8 = 24, and 5% = 4 -> 35% = 24 + 4 = 28.
2. Percent Equation Method: Write 35% as 0.35 (or 35/100) and multiply by the Whole: Part = 0.35 × 80 = 28.

Block Connection (ALGB-01): Way A (Add inside first): 6 × 14 = 84. Way B (Split into 2 rooms): 6×10 (60) + 6×4 (24) = 84!

SC-6 Keystrokes: 6 ENTER 10 × 6 ENTER 4 × +

Answer: 84 (Matches 6 × 14 = 84) *Floor 2 holds Room 1 (60) while Floor 1 holds Room..*

SECTION 4.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Double Number Line, 10% Mental Benchmark & Percent Equation (Pa..

1 Big Ideas Method: 10% Benchmark / Percent Equation
Compute 7 × (20 + 3) both ways on your SC-6. Write the two partial products.
Work:
Answer: _____

2 Big Ideas Method: 10% Benchmark / Percent Equation
Build 4 rows of (x + 5) and write the distributed expression for 4(x + 5).
Work:
Answer: _____

3 Big Ideas Method: 10% Benchmark / Percent Equation
Use the Distributive Property to expand 6(3x + 4).
Work:
Answer: _____

4 Big Ideas Method: 10% Benchmark / Percent Equation
Expand 8(2y - 5). Watch the subtraction sign!
Work:
Answer: _____

5 Big Ideas Method: Part = Percent × Whole
Mental Math Split: Compute 9 × 43 by splitting 43 into (40 + 3).
Work:
Answer: _____

6 Big Ideas Method: Part = Percent × Whole
Distribute and combine like terms: 3(4x + 2) + 5x.
Work:
Answer: _____

7 Big Ideas Method: Part = Percent × Whole
Distribute and combine: 4(3n + 5) - 7.
Work:
Answer: _____

8 Big Ideas Method: Part = Percent × Whole
Spot the Glitch: Mia wrote 5(x + 6) = 5x + 6. What did she forget to multiply by 5?
Work:
Answer: _____

9 Application & 2-Way Check: Tip, Tax & Discount Part
Simplify 5(2x + 3) + 4x AND prove your answer matches the original when x = 2!
Work:
Answer: _____

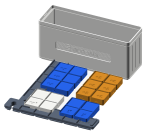
10 Application & 2-Way Check: Tip, Tax & Discount Part
6 friends each buy a burger ($x) and $3.50 fries: 6(x + 3.50). Expand it and evaluate for x = $6.50!
Work:
Answer: _____

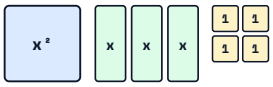
Section 4.4: Factoring Expressions with GCF — Reverse the Rectangle

Big Ideas Method: Percent Proportion (Part / Whole = p / 100) & Tape Diagram to Find the Whole (Whole = Part ÷ %) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
'Factoring' sounds hard, but it is just running Section 1.5's Distributive Property in reverse using Section 1.5's [GCF] key! To factor 18x + 24, press 18 [ENTER] 24 [GCF] -> 6 (that's the number of equal rows!). Divide 18x and 24 by 6 to get 6(3x + 4)!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA & PRIME STUDIO)





Sort by Shape:
x²-Flats | x-Rods | 1-Cubes

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA & PRIME STUDIO):
Step 1: Open Kit COUNT-01 / TENF-01 and build a 4-quarter Percent Tape (each block = 25%, total = 100%).
Step 2: Place 18 onto the three shaded 25% blocks (75% = 3/4 of the tape), so 1 block (25%) = 18 ÷ 3 = 6.
Step 3: Multiply 4 blocks × 6 per block = 24 for the 100% Whole!

WORKED EXAMPLE — EXAMPLE 1: FACTORING NUMBERS & ALGEBRA WITH [GCF]

Problem: Factor 18x + 24 completely using the [GCF] key.

Big Ideas Textbook Method (Percent Proportion (Part / Whole = p / 100) & Tape Diagram to Find the Wh..
1. Set up the Percent Proportion: Part / Whole = Percent / 100 -> 18 / w = 75 / 100.
2. Or use the Percent Equation: 18 = 0.75 × w.
3. Divide both sides by 0.75 to isolate the Whole: w = 18 ÷ 0.75 = 24.

Block Connection (ALGB-01): Find the largest number of equal rows: 18 [ENTER] 24 [GCF] -> 6 rows! Each of the 6 rows has 18÷6 = 3 x-rods and 24÷6 = 4 cubes.

SC-6 Keystrokes: 18 | ENTER | 24 | GCF

Answer: 6(3x + 4) *Check by distributing back: 6 × 3x = 18x and 6 × 4 =..*

SECTION 4.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Percent Proportion (Part / Whole = p / 100) & Tape Diagram to F..

1 Big Ideas Method: Percent Proportion (a/w = p/100)
Factor 12 + 20 into the form a(b + c) using [GCF].
Work:
Answer: _____

2 Big Ideas Method: Percent Proportion (a/w = p/100)
Factor 8x + 12 using [GCF].
Work:
Answer: _____

3 Big Ideas Method: Percent Proportion (a/w = p/100)
Factor 15y - 25 using [GCF].
Work:
Answer: _____

4 Big Ideas Method: Percent Proportion (a/w = p/100)
Factor 24x + 36 completely using [GCF].
Work:
Answer: _____

5 Big Ideas Method: Find Whole (Part ÷ Percent)
Factor 32n - 48 completely using [GCF].
Work:
Answer: _____

6 Big Ideas Method: Find Whole (Part ÷ Percent)
Factor 42a + 14 completely. (Careful: what is 14 ÷ 14?)
Work:
Answer: _____

7 Big Ideas Method: Find Whole (Part ÷ Percent)
Factor 36m - 60 completely using [GCF].
Work:
Answer: _____

8 Big Ideas Method: Find Whole (Part ÷ Percent)
Spot the Glitch: Sam factored 24x + 36 as 6(4x + 6). Why isn't it factored completely?
Work:
Answer: _____

9 Application & 2-Way Check: Reverse Discount / Goal Whole
A rectangular banner has area (28x + 42) sq in and height 14 in. Factor with [GCF] to find its width!
Work:
Answer: _____

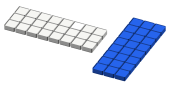
10 Application & 2-Way Check: Reverse Discount / Goal Whole
A square garden has perimeter (36x + 48) ft. Factor out 4 to find the length of 1 side!
Work:
Answer: _____

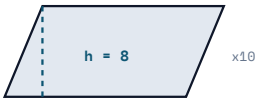
Section 5.1: Areas of Parallelograms & Rectangles ([f] [RECT])

Big Ideas Method: Substituting Values for Variables in Expressions & Applying Order of Operations Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Textbooks always label a trap number on parallelograms: Base b, straight vertical Height h, AND the slanted side s. If you chop the corner triangle off one end and slide it to the other, every parallelogram turns into a straight b × h Rectangle! Cross out the slanted side and press b [ENTER] h [f] [RECT] ([f][7])!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and place 3 green x-rods and 5 yellow unit cubes on the stage for 3x + 5.

Step 2: To evaluate at x = 4, replace each green x-rod with 4 unit cubes (3 groups of 4 = 12 cubes).

Step 3: Combine the 12 cubes with the 5 constant cubes to count 12 + 5 = 17.

WORKED EXAMPLE — EXAMPLE 1: DODGE THE SLANTED TRAP WITH [F] [RECT]

Problem: Find the area of a parallelogram with base b = 12 cm, vertical height h = 8 cm, and slanted side 10 cm.

Big Ideas Textbook Method (Substituting Values for Variables in Expressions & Applying Order of Ope..

1. Write the algebraic expression: 3x + 5.
2. Substitute 4 for the variable x using parentheses: 3(4) + 5.
3. Follow the Order of Operations (multiply before adding): 12 + 5 = 17.

Block Connection (COUNT-01): Cross out the slanted 10 cm trap! Push Base 12 onto Y, vertical Height 8 onto X, and press [f] [RECT] ([f][7]).

SC-6 Keystrokes: `12` `ENTER` `8` `f` `RECT`

Answer: 96 cm²

[f] [RECT] multiplies Base (Y) × Height (X) -> 96 cm²!

SECTION 5.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Substituting Values for Variables in Expressions & Applying Ord..

1 Big Ideas Method: Variable Substitution

Find the area of a parallelogram with base b = 15 m, vertical height h = 9 m, and slanted side 11 m.

Work:

Answer: _____

2 Big Ideas Method: Variable Substitution

Find the area of a rectangle measuring 18 ft by 12 ft using [f] [RECT].

Work:

Answer: _____

3 Big Ideas Method: Variable Substitution

Find the area of a parallelogram with b = 8.5 m and vertical height h = 6.4 m.

Work:

Answer: _____

4 Big Ideas Method: Variable Substitution

Find the area of a square tile with side 13 cm.

Work:

Answer: _____

5 Big Ideas Method: Two-Variable & Power Evaluation

A parallelogram has base b = 6 1/2 in and height h = 4 in. Find its area.

Work:

Answer: _____

6 Big Ideas Method: Two-Variable & Power Evaluation

Missing Height: A parallelogram has Area = 108 cm² and base b = 12 cm. Find h.

Work:

Answer: _____

7 Big Ideas Method: Two-Variable & Power Evaluation

Missing Base: A parallelogram has Area = 135 ft² and vertical height h = 7.5 ft. Find b.

Work:

Answer: _____

8 Big Ideas Method: Two-Variable & Power Evaluation

Spot the Glitch: A parallelogram has b = 10, vertical h = 6, slanted side = 7.5. Sam got 75. Fix it!

Work:

Answer: _____

9 Application & 2-Way Check: Formula Substitution

A slanted parking space has base 9 ft and straight length 18.5 ft. How much asphalt covers 4 spaces?

Work:

Answer: _____

10 Application & 2-Way Check: Formula Substitution

A rectangular patio (15 ft × 12 ft) has a parallelogram planter (b=4 ft, h=3 ft) inside. Find the remaining patio area!

Work:

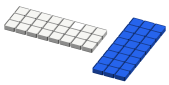
Answer: _____

Section 5.2: Areas of Triangles — Half a Rectangle ([g] [TRI])

Big Ideas Method: Translating Verbal Phrases into Algebraic Expressions & Identifying Terms, Coefficients, Constants **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
The #1 geometry test mistake is multiplying b × h for a triangle and forgetting to divide by 2! Look at key 7 on your SC-6: [f] [RECT] is b × h, and right next to it, [g] [TRI] ([g][7]) computes ½ × b × h in a single click so you never forget the ½!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





[g] [TRI] = ½ × b × h

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and place 2 green x-rods on the stage to model 'twice a number x' (2x).
Step 2: For '7 less than twice a number x', start with 2x and take 7 away: 2x - 7 (never 7 - 2x!).
Step 3: Point to the parts on the stage: Terms = 2x and -7; Coefficient of x = 2; Constant = -7.

WORKED EXAMPLE — EXAMPLE 1: ONE-CLICK TRIANGLE AREA WITH [G] [TRI]

Problem: Find the area of a triangle with base b = 10 cm and height h = 6 cm.

Big Ideas Textbook Method (Translating Verbal Phrases into Algebraic Expressions & Identifying Term..
1. Identify operation words: 'sum' (+), 'difference' (-), 'product' (×), 'quotient' (÷).
2. Watch turnaround words ('less than' or 'subtracted from'): '7 less than 2x' translates to 2x - 7.
3. Identify parts of 5x + 3y + 8: Terms = 5x, 3y, 8; Coefficients = 5, 3; Constant = 8.

Block Connection (COUNT-01): A 10 × 6 rectangle has area 60 cm². Cutting it diagonally in half gives two identical triangles of 30 cm² each!

SC-6 Keystrokes: `10` `ENTER` `6` `g` `TRI`

Answer: 30 cm²

[g] [TRI] ([g][7]) multiplies Y × X and divides by 2 auto..

SECTION 5.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Translating Verbal Phrases into Algebraic Expressions & Identif..

1 Big Ideas Method: Verbal-to-Algebraic Translation
Find the area of a triangle with base b = 14 ft and height h = 18 ft using [g] [TRI].

Work:

Answer: _____

2 Big Ideas Method: Verbal-to-Algebraic Translation
Find the area of a right triangle with base 9 ft, vertical height 4 ft, and slanted hypotenuse 9.85 ft.

Work:

Answer: _____

3 Big Ideas Method: Verbal-to-Algebraic Translation
Find the area of a triangle with b = 16 cm and h = 11 cm.

Work:

Answer: _____

4 Big Ideas Method: Verbal-to-Algebraic Translation
Find the area of a triangle with decimal dimensions: b = 12.5 m, h = 8.4 m.

Work:

Answer: _____

5 Big Ideas Method: Turnaround Phrase ('Less Than')
Find the area of a triangle with fractional dimensions: b = 4 1/2 in, h = 6 in.

Work:

Answer: _____

6 Big Ideas Method: Turnaround Phrase ('Less Than')
Missing Height: A triangle has Area = 54 cm² and base b = 12 cm. (Double 54 first, then ÷ 12!)

Work:

Answer: _____

7 Big Ideas Method: Turnaround Phrase ('Less Than')
Missing Base: A triangle has Area = 84 m² and height h = 14 m. Find base b.

Work:

Answer: _____

8 Big Ideas Method: Turnaround Phrase ('Less Than')
Spot the Glitch: Leo found the area of a b=8, h=5 triangle as 40. What did he calculate instead?

Work:

Answer: _____

9 Application & 2-Way Check: Terms, Coefficients & Modeling
A sailboat has a main sail (b=12 ft, h=25 ft) and jib sail (b=8 ft, h=15 ft). Find total sail area 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Terms, Coefficients & Modeling
A triangular gable window has Area = 36 ft² and base 9 ft. Find its height and check with [g] [TRI]!

Work:

Answer: _____

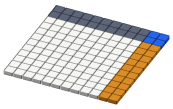
Section 5.3: Trapezoids & Composite Figures with the [Σ+] Running Ledger

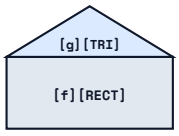
Big Ideas Method: Commutative (Reorder) & Associative (Regroup) Properties of Addition and Multiplication Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Composite figures (weird shapes made of rectangles and triangles) are easy when you use the SC-6 [Σ+] running accumulator! Clear memory with [g] [CLΣ] ([g][÷]). Compute each simple piece ([f] [RECT] or [g] [TRI]) and tap [Σ+] to toss it in the bucket (use [f] [Σ-] to subtract a cutout hole), then read the grand total with [f] [Σx] ([f][N])!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





Sum chunks with [Σ+]!

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit ALGB-01 / COUNT-01 and place a green x-rod bracketed with 17 unit cubes, plus 23 unit cubes: (x + 17) + 23.

Step 2: Slide the grouping bracket so the unit cubes sit together: x + (17 + 23) (Associative Property of Addition).

Step 3: Combine the unit cubes into 40 to read the simplified expression x + 40.

WORKED EXAMPLE — EXAMPLE 1: HOUSE WALL COMPOSITE (RECTANGLE + TRIANGLE VIA [Σ+])

Problem: Find the total area of a 12 ft × 8 ft rectangular wall topped by a roof triangle (b = 12 ft, h = 5 ft).

Big Ideas Textbook Method (Commutative (Reorder) & Associative (Regroup) Properties of Addition a..

1. Use the Commutative Property to reorder compatible numbers next to each other.
2. Use the Associative Property to regroup the constants: (x + 17) + 23 = x + (17 + 23).
3. Add or multiply the grouped constants to simplify: x + 40.

Block Connection (COUNT-01): Clear stats ([g] [CLΣ]). Piece 1: 12 [ENTER] 8 [f] [RECT] [Σ+] (96, n=1). Piece 2: 12 [ENTER] 5 [g] [TRI] [Σ+] (30, n=2). Read [f] [Σx]!

SC-6 Keystrokes: g CLΣ 12 ENTER 8 f RECT Σ+ 12 ENTER 5 g TRI Σ+ f Σx

Answer: 126 ft²

The screen counts n = 1, n = 2 so you never forget a..

SECTION 5.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Commutative (Reorder) & Associative (Regroup) Properties of Add..

1 Big Ideas Method: Commutative / Associative Mental Math

Find the area of a trapezoid with b1 = 10 cm, b2 = 14 cm, and h = 6 cm using (b1+b2) [ENTER] h [g] [TRI].

Work:

Answer: _____

2 Big Ideas Method: Commutative / Associative Mental Math

Find the area of a trapezoid with b1 = 18 ft, b2 = 12 ft, and h = 9 ft.

Work:

Answer: _____

3 Big Ideas Method: Commutative / Associative Mental Math

L-Shape Room: Split into Room A (10 × 6) and Room B (8 × 4). Sum with [Σ+] and [f] [Σx].

Work:

Answer: _____

4 Big Ideas Method: Commutative / Associative Mental Math

Banner Arrow: A 15 × 6 rectangle attached to a triangle (b = 10, h = 6). Find total area.

Work:

Answer: _____

5 Big Ideas Method: Simplifying Algebraic Groupings

Find the area of a trapezoid with decimal bases b1 = 8.5 m, b2 = 5.5 m, and h = 4.2 m.

Work:

Answer: _____

6 Big Ideas Method: Simplifying Algebraic Groupings

Cutout Frame: A 10 in × 8 in frame has a 6 in × 4 in hole cut out. Use [Σ+] and [f] [Σ-] to find the border area!

Work:

Answer: _____

7 Big Ideas Method: Simplifying Algebraic Groupings

Missing Trapezoid Height: A trapezoid has Area = 90 cm² and bases b1 = 11 cm, b2 = 7 cm. Find h.

Work:

Answer: _____

8 Big Ideas Method: Simplifying Algebraic Groupings

A mini-golf hole is a 12 × 5 ft rectangle plus a trapezoid (b1=5, b2=9, h=4). Find total area.

Work:

Answer: _____

9 Application & 2-Way Check: Property Justification

Find the area of a trapezoid (b1=10, b2=16, h=8) both with the trapezoid formula AND by splitting into a 10×8 rect + 6×8 triangle!

Work:

Answer: _____

10 Application & 2-Way Check: Property Justification

A 20 ft × 15 ft lawn has a triangular flowerbed (b=8 ft, h=6 ft) and square shed (5 ft × 5 ft). Find grass area!

Work:

Answer: _____

Section 5.4: Coordinate Polygons & Planetary Clock Base-60 Time ([CLK])

Big Ideas Method: Distributive Property Area Model — Expanding a(b + c) = ab + ac & Factoring Out the GCF Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

On a coordinate grid, find rectangle side lengths by subtracting the x-coordinates and y-coordinates. And when adding/subtracting hours and minutes, remember time is Base-60! Press [CLK] on the SC-6 to enter H.MM mode (1.45 [ENTER] 1.35 [+] -> 3.20), or use [f] [->HR] ([f][CLK]) to turn 3 hr 15 min (3.15) into 3.25 decimal hours!

BLOCK KIT & STEP-BY-STEP GUIDE (TIME-01: PLANETARY GEARED CLOCK)





60 min = 1 hr ([CLK])

HOW TO BUILD IT WITH TIME-01 (PLANETARY GEARED CLOCK):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and build 3 identical rows of (1 green x-rod + 4 yellow unit cubes) for 3(x + 4).

Step 2: Count the total green x-rods across all 3 rows (3 × x = 3x) and the total yellow cubes (3 × 4 = 12) -> 3x + 12.

Step 3: Reverse the process to factor 12x + 18: divide into 6 equal rows (GCF = 6) of (2x + 3) -> 6(2x + 3).

WORKED EXAMPLE — EXAMPLE 1: COORDINATE RECTANGLE PERIMETER & AREA

Problem: A rectangle has vertices (2, 3), (9, 3), (9, 8), and (2, 8). Find its width, height, perimeter, and area.

Big Ideas Textbook Method (Distributive Property Area Model — Expanding a(b + c) = ab + ac & Factor..

1. To expand 3(x + 4), multiply the outside factor 3 by BOTH inside terms: 3(x) + 3(4) = 3x + 12.
2. To factor 12x + 18, find the GCF of the coefficients 12 and 18: GCF(12, 18) = 6.
3. Divide each term by 6 and write inside parentheses: 6(2x + 3).

Block Connection (TIME-01): Width = 9 - 2 = 7 units. Height = 8 - 3 = 5 units. Area = 7 × 5 = 35 sq units; Perimeter = 2(7 + 5) = 24 units.

SC-6 Keystrokes: 9 ENTER 2 - 8 ENTER 3 - f RECT

Answer: Width = 7, Height = 5, Perim *Subtract x-coords for horizontal width; subtract y-coor..*

SECTION 5.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Distributive Property Area Model — Expanding a(b + c) = ab + ac..

1 Big Ideas Method: Distributive Expansion a(b+c)

Find the distance between (3, 4) and (11, 4) on a coordinate grid.

Work:

Answer: _____

2 Big Ideas Method: Distributive Expansion a(b+c)

A rectangle has corners (1, 2), (10, 2), (10, 8), (1, 8). Find its area.

Work:

Answer: _____

3 Big Ideas Method: Distributive Expansion a(b+c)

In [CLK] mode, add 1 hr 45 min + 1 hr 35 min (1.45 [ENTER] 1.35 [+]).

Work:

Answer: _____

4 Big Ideas Method: Distributive Expansion a(b+c)

Convert 2 hr 30 min (2.30) to decimal hours using [f] [->HR].

Work:

Answer: _____

5 Big Ideas Method: Factoring Out the GCF

Convert 4 hr 45 min (4.45) to decimal hours using [f] [->HR].

Work:

Answer: _____

6 Big Ideas Method: Factoring Out the GCF

Convert 1.75 decimal hours back into Hours & Minutes using [g] [->HMS].

Work:

Answer: _____

7 Big Ideas Method: Factoring Out the GCF

In [CLK] mode, find elapsed time from 8:50 AM to 11:25 AM (11.25 [ENTER] 8.50 [-]).

Work:

Answer: _____

8 Big Ideas Method: Factoring Out the GCF

A right triangle on a grid has vertices (2, 1), (10, 1), and (2, 7). Find its area!

Work:

Answer: _____

9 Application & 2-Way Check: 2-Way Area Model Equivalence

You tutor for 3 hr 15 min (3.15) at $16/hr. Convert to decimal hours with [f] [->HR] and find your pay 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: 2-Way Area Model Equivalence

Coordinate Park: Vertices (2, 2), (14, 2), (14, 11), (2, 11) where 1 unit = 10 yd. Find perimeter and area!

Work:

Answer: _____

Section 6.1: Positive & Negative Integers — Zero Pairs & The [+/-] Sign Flipper

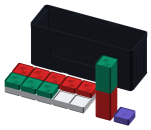
Big Ideas Method: Positive & Negative Integers on a Number Line & Finding Opposites Across 0

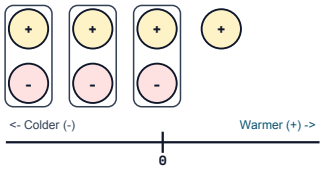
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CONCEPT SUMMARY & TEXTBOOK CONNECTION:

On regular calculators, mixing up the minus key and negative key causes SYNTAX ERROR. On the SC-6, [-] is the two-number subtraction action, while [+/-] flips the sign of the number on Floor 1 (X) between Positive (+) and Negative (-)—just like flipping a counter from Yellow (+) to Red (-)!

BLOCK KIT & STEP-BY-STEP GUIDE (BASE-10: TEN-FRAME ZERO-PAIR COUNTERS)





HOW TO BUILD IT WITH BASE-10 (TEN-FRAME ZERO-PAIR COUNTERS):

Step 1: Open Kit RACK-01 (Number Rack & Zero-Pair Studio) and set the center divider as Zero (0 / Sea Level).

Step 2: Slide 7 positive beads to the right (+7) and 7 negative beads to the left (-7) at the exact same distance from 0.

Step 3: Flip a number across 0 once for its opposite (-7), and flip twice to return to the original number: -(-7) = +7.

WORKED EXAMPLE — EXAMPLE 1: OPPOSITES & ZERO-PAIR CANCELLATION WITH [+/-]

Problem: Enter -15 on the SC-6, find its opposite, and verify that -15 + 15 = 0.

Big Ideas Textbook Method (Positive & Negative Integers on a Number Line & Finding Opposites Acro..

1. Identify words for positive (+) integers (gain, above, deposit) and negative (-) integers (loss, below, withdrawal).
2. The opposite of a number is the same distance from 0 on the opposite side of 0: Opposite of 7 is -7.
3. The opposite of the opposite of a number is the number itself: -(-7) = 7.

Block Connection (BASE-10): 15 Red (-1) counters and 15 Yellow (+1) counters pair up 1-for-1 into 15 Zero Pairs (= 0).

SC-6 Keystrokes: `15` `+/-` `ENTER` `15` `+`

Answer: Opposite of -15 is +15; -15.. *Type the digits 15 first, THEN tap [+/-] to make it -15!*

SECTION 6.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Positive & Negative Integers on a Number Line & Finding Opposit..

1 Big Ideas Method: Integer Modeling & Sign

Write the integer for '25 feet below sea level' and find its opposite using [+/-].

Work:

Answer: _____

2 Big Ideas Method: Integer Modeling & Sign

Find the opposite of -42, and then the opposite of the opposite (-[-(-42)]).

Work:

Answer: _____

3 Big Ideas Method: Integer Modeling & Sign

Zero-Pair Cancellation: A submarine is at -18 m and rises +18 m. Where is it now?

Work:

Answer: _____

4 Big Ideas Method: Integer Modeling & Sign

Combine 6 Yellow (+1) counters and 6 Red (-1) counters. What is the value?

Work:

Answer: _____

5 Big Ideas Method: Opposites & Opposite-of-Opposite

A hiker starts at -120 ft in a canyon and climbs +120 ft. What is her elevation?

Work:

Answer: _____

6 Big Ideas Method: Opposites & Opposite-of-Opposite

What integer is its own opposite? Test 0 [+/-] on your SC-6!

Work:

Answer: _____

7 Big Ideas Method: Opposites & Opposite-of-Opposite

Temperature starts at 0°F, drops 12°F (12 [+/-]), and rises 19°F. Find the temperature.

Work:

Answer: _____

8 Big Ideas Method: Opposites & Opposite-of-Opposite

Golf Par: -3 means 3 strokes under par. If you shoot -3 on Saturday and +3 on Sunday, what is your total?

Work:

Answer: _____

9 Application & 2-Way Check: Real-World Zero Reference

Elevator starts at Floor 0, goes up 14, down 20 (20 [+/-]), and up 9. Find the floor 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Real-World Zero Reference

Game XP: You gain 40 XP, lose 65 XP, and gain 25 XP. Did you break even? Prove 2 ways!

Work:

Answer: _____

Section 6.2: Comparing & Ordering Integers and Rational Numbers

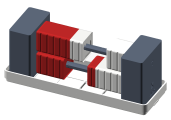
Big Ideas Method: Comparing & Ordering Integers and Rational Numbers on a Horizontal/Vertical Number Line

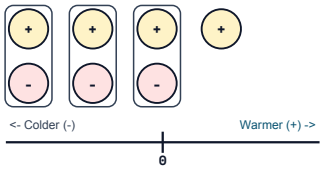
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CONCEPT SUMMARY & TEXTBOOK CONNECTION:

When kids see -18 and -5, their instinct says -18 is bigger because 18 > 5. Think of temperature: -5°F is WARMER than -18°F! On a horizontal number line ([f] [LINE]), further to the RIGHT is always greater. Test any inequality on the SC-6 with [f] [>] ([f][≠]) and [g] [<] ([g][≠]): 1 = TRUE, 0 = FALSE!

BLOCK KIT & STEP-BY-STEP GUIDE (RACK-01: NUMBER RACK & NUMBER LINE)





HOW TO BUILD IT WITH RACK-01 (NUMBER RACK & NUMBER LINE):

Step 1: Open Kit RACK-01 (Number Rack) and mark 0 at the center, with negative numbers extending left and positive right.

Step 2: Locate -4 (4 steps left of 0) and -9 (9 steps left of 0).

Step 3: Use the Number Line Rule: further RIGHT (or higher up on a thermometer) is ALWAYS greater, so -4 > -9!

WORKED EXAMPLE — EXAMPLE 1: TESTING NEGATIVE INEQUALITIES WITH [F] [>] & [G]..

Problem: Compare -5 and -18. Which one is greater?

Big Ideas Textbook Method (Comparing & Ordering Integers and Rational Numbers on a Horizontal/Ve..

1. Plot both numbers (-4 and -9) on a horizontal number line.
2. Because -4 is to the right of -9 (closer to 0 and warmer), -4 is greater than -9 (-4 > -9).
3. To order a list from least to greatest, read the plotted points from left to right.

Block Connection (RACK-01): On the number line, -5 is 5 steps left of 0, while -18 is 18 steps left of 0. Since -5 is to the right of -18, -5 > -18!

SC-6 Keystrokes: `5` `+/-` `ENTER` `18` `+/-` `f` `>`

Answer: -5 > -18 (SC-6 outputs 1 = T.. *If you test [g] [<], the SC-6 outputs 0 (FALSE)!*

SECTION 6.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Comparing & Ordering Integers and Rational Numbers on a Horizon..

1 Big Ideas Method: Number Line Inequality (< or >)

Which temperature is warmer (greater): -12°C or -5°C? Test with [g] [<].

Work:

Answer: _____

2 Big Ideas Method: Number Line Inequality (< or >)

Compare -9 and -2 using > or <.

Work:

Answer: _____

3 Big Ideas Method: Number Line Inequality (< or >)

Compare -15 and 0 using > or <.

Work:

Answer: _____

4 Big Ideas Method: Number Line Inequality (< or >)

Order the integers from least to greatest: 4, -7, 0, -2, -10.

Work:

Answer: _____

5 Big Ideas Method: Ordering Negative Decimals/Fractions

Compare -3/4 and -0.6. (Hint: -3/4 = -0.75. Which is further right?)

Work:

Answer: _____

6 Big Ideas Method: Ordering Negative Decimals/Fractions

Compare -2 1/2 and -2.3 using > or <.

Work:

Answer: _____

7 Big Ideas Method: Ordering Negative Decimals/Fractions

Compare -5/8 and -7/8.

Work:

Answer: _____

8 Big Ideas Method: Ordering Negative Decimals/Fractions

Spot the Glitch: Leo says -4.5 > -4.1 because 4.5 > 4.1. Explain why -4.1 is actually greater!

Work:

Answer: _____

9 Application & 2-Way Check: Elevation & Thermometer Ranking

Order -1.5, 1 1/4, -1 3/4, and -0.5 from least to greatest. Verify both as decimals AND with [g] [<]!

Work:

Answer: _____

10 Application & 2-Way Check: Elevation & Thermometer Ranking

Three divers are at -14.8 m, -14 1/2 m, and -15.2 m. Who is deepest (least) and who is shallowest (greatest)?

Work:

Answer: _____

Section 6.3: Absolute Value — Distance From Zero Is Always Positive ([g] [|x|])

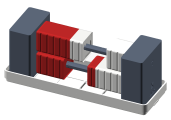
Big Ideas Method: Absolute Value |x| as Distance from Zero (Always Non-Negative) & Evaluating |x| Expressions

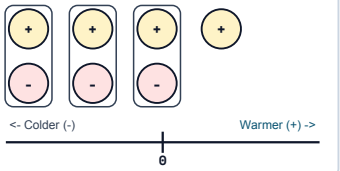
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CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Students often confuse Opposite ([+/-]) with Absolute Value ([g] [|x|], [g][+/-]). Opposite flips the sign (+9 -> -9, -9 -> +9). Absolute Value asks 'How many steps from 0?' Since you can't walk a negative number of steps, [g] [|x|] is ALWAYS positive (or 0)!

BLOCK KIT & STEP-BY-STEP GUIDE (RACK-01: NUMBER RACK & ZERO-PAIR STUDIO)





HOW TO BUILD IT WITH RACK-01 (NUMBER RACK & ZERO-PAIR STUDIO):

Step 1: Open Kit RACK-01 (Number Rack) and place a marker at -15 on the left side of 0.

Step 2: Count the physical steps from -15 back to 0: distance is 15 steps, so |-15| = +15.

Step 3: To evaluate 50 - |-18|, first find the distance |-18| = 18, then subtract 50 - 18 = 32.

WORKED EXAMPLE — EXAMPLE 1: OPPOSITE ([+/-]) VS. ABSOLUTE VALUE ([G] [|X|])

Problem: Find |-14|, |14|, and -|-14| on your SC-6.

Big Ideas Textbook Method (Absolute Value |x| as Distance from Zero (Always Non-Negative) & Evalua..

1. Absolute value |x| measures distance from 0 on a number line, so |x| is never negative: |-15| = 15.
2. Treat absolute value bars like grouping symbols—evaluate inside | | first: |-27| + |13| = 27 + 13 = 40.
3. A negative sign outside the bars applies after finding the distance: -|-24| = -24.

Block Connection (RACK-01): -14 is 14 steps from 0, so |-14| = 14. In -|-14|, find |-14| = 14 first, then apply the outside negative sign -> -14!

SC-6 Keystrokes: `14` `+/-` `g` `|x|`

Answer: |-14| = 14, |14| = 14, -|-14 *[g] [|x|] ([g][+/-]) makes any number on Floor 1 positive!*

SECTION 6.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Absolute Value |x| as Distance from Zero (Always Non-Negative)..

1 Big Ideas Method: Distance from Zero |x|

Find |-27| using [g] [|x|].

Work:

Answer: _____

2 Big Ideas Method: Distance from Zero |x|

Evaluate |-27| + |13|.

Work:

Answer: _____

3 Big Ideas Method: Distance from Zero |x|

Evaluate |-4.8| + |-3.2|.

Work:

Answer: _____

4 Big Ideas Method: Distance from Zero |x|

Evaluate |-35| - |-19|.

Work:

Answer: _____

5 Big Ideas Method: Absolute Value Expressions

Compare |-12| and |9| using > or <. (Which is further from 0?)

Work:

Answer: _____

6 Big Ideas Method: Absolute Value Expressions

Evaluate -|-24|. (Find |-24| first, then press [+/-] for the outside minus!)

Work:

Answer: _____

7 Big Ideas Method: Absolute Value Expressions

Evaluate 50 - |-18|. Find |-18| first, then subtract from 50.

Work:

Answer: _____

8 Big Ideas Method: Absolute Value Expressions

Order from least to greatest: |-8|, -6, |-3|, 0, -|-9|.

Work:

Answer: _____

9 Application & 2-Way Check: Magnitude (Debt/Depth) vs. Value

A diver is at -32 ft and a seagull is at +25 ft. Which is closer to sea level (0), and by how many feet?

Work:

Answer: _____

10 Application & 2-Way Check: Magnitude (Debt/Depth) vs. Value

Two accounts have balances of -$45 and -$28. Which account has the greater debt (|-45| vs |-28|), and which has the greater balance?

Work:

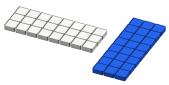
Answer: _____

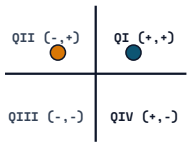
Section 6.4: The 4-Quadrant Coordinate Plane & Axis Reflections

Big Ideas Method: Plotting (x, y) in All Four Quadrants (I, II, III, IV) & Reflecting Across the x- and y-Axes Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
To never mix up (x, y), use the Elevator Rule: walk ACROSS the lobby first (x: right +, left -), then ride UP/DOWN the elevator (y: up +, down -)! Reflecting across the y-axis flips the x-sign ([+/-]); reflecting across the x-axis flips the y-sign ([+/-])!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: 4X4 GRIDFINITY BASEPLATE & TIL..





HOW TO BUILD IT WITH COUNT-01 (4X4 GRIDFINITY BASEPLATE & TILES):

Step 1: Open Kit COUNT-01 (4x4 Gridfinity Baseplate) and divide the grid into 4 quadrants around the center origin (0, 0).

Step 2: To plot (-4, 3), walk 4 columns LEFT along the x-axis, then climb 3 rows UP into Quadrant II.

Step 3: Reflect across the y-axis by flipping the x-sign to (+4, 3); reflect across the x-axis by flipping the y-sign to (-4, -3).

WORKED EXAMPLE — EXAMPLE 1: NAMING QUADRANTS I, II, III, AND IV

Problem: Identify the quadrant for A(-6, 5), B(4, -7), and C(-3, -8).

Big Ideas Textbook Method (Plotting (x, y) in All Four Quadrants (I, II, III, IV) & Reflecting Across the x-..

1. Quadrant Signs: QI (+, +), QII (-, +), QIII (-, -), QIV (+, -).
2. Reflect across the x-axis: keep x the same and take the opposite of y -> (x, -y).
3. Reflect across the y-axis: take the opposite of x and keep y the same -> (-x, y).

Block Connection (COUNT-01): A(-, +) goes left 6, up 5 -> Quad II. B(+, -) goes right 4, down 7 -> Quad IV. C(-, -) goes left 3, down 8 -> Quad III.

SC-6 Keystrokes: `6` `+/-` `ENTER` `5`

Answer: A: Quad II, B: Quad IV, C: Q.. *QI: (+,+), QII: (-,+), QIII: (-,-), QIV: (+,-)!*

SECTION 6.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Plotting (x, y) in All Four Quadrants (I, II, III, IV) & Reflec..

1 Big Ideas Method: Quadrant Identification (I-IV)
Name the Quadrant (I, II, III, or IV) for point (7, -4).

Work:

Answer: _____

2 Big Ideas Method: Quadrant Identification (I-IV)
Name the Quadrant for point (-9, -2).

Work:

Answer: _____

3 Big Ideas Method: Quadrant Identification (I-IV)
Name the Quadrant for point (-5, 11).

Work:

Answer: _____

4 Big Ideas Method: Quadrant Identification (I-IV)
Reflect (6, 9) across the y-axis. (Flip the x-coordinate with [+/-]!)

Work:

Answer: _____

5 Big Ideas Method: x-Axis & y-Axis Reflections
Reflect (6, 9) across the x-axis. (Flip the y-coordinate with [+/-]!)

Work:

Answer: _____

6 Big Ideas Method: x-Axis & y-Axis Reflections
Reflect (-3, -7) across the x-axis.

Work:

Answer: _____

7 Big Ideas Method: x-Axis & y-Axis Reflections
Reflect (-8, 4) across the y-axis, and then across the x-axis.

Work:

Answer: _____

8 Big Ideas Method: x-Axis & y-Axis Reflections
On the Fence: Which axis does (0, -9) sit on, and which axis does (-7, 0) sit on?

Work:

Answer: _____

9 Application & 2-Way Check: Coordinate Grid Navigation
Start at (4, -5). Walk 7 units left and 8 units up. Where do you land, and in which quadrant?

Work:

Answer: _____

10 Application & 2-Way Check: Coordinate Grid Navigation
Points A(-4, 2), B(4, 2), and C(4, -5) are 3 corners of a rectangle. Find the 4th corner D!

Work:

Answer: _____

Section 6.5: Distance Between Points on the Coordinate Plane ([g] [|x|])

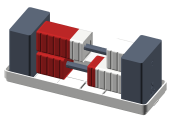
Big Ideas Method: Finding Horizontal & Vertical Distances on the Coordinate Plane Using Absolute Value

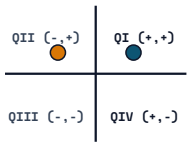
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CONCEPT SUMMARY & TEXTBOOK CONNECTION:

When two points share a coordinate like (-6, 4) and (9, 4), how far apart are they? If they are in different quadrants (cross an axis), add their distances to the axis (|-6| + |9| = 15). Even faster: on the SC-6, subtract ANY two coordinates and press [g] [|x|] (-6 - 9 = -15 -> |-15| = 15)!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: BASEPLATE & ZERO-PAIR STUDIO)





HOW TO BUILD IT WITH COUNT-01 (BASEPLATE & ZERO-PAIR STUDIO):

Step 1: Open Kit COUNT-01 / RACK-01 and place tiles at (-6, 4) in Quadrant II and (9, 4) in Quadrant I.

Step 2: Count the steps from (-6, 4) to the y-axis (|-6| = 6 steps) plus the steps from the y-axis to (9, 4) (|9| = 9 steps).

Step 3: Add the two distances across the axis: 6 + 9 = 15 units (or compute |-6 - 9| = |-15| = 15).

WORKED EXAMPLE — EXAMPLE 1: CROSS-AXIS DISTANCE VS. SAME-QUADRANT DISTANCE

Problem: Find the distance between (-6, 4) and (9, 4), and between (-12, -3) and (-4, -3).

Big Ideas Textbook Method (Finding Horizontal & Vertical Distances on the Coordinate Plane Using Ab..

1. Confirm the points share a coordinate: (-6, 4) and (9, 4) share y = 4 (horizontal segment).
2. Different Quadrants (opposite signs): Add the absolute values: |-6| + |9| = 6 + 9 = 15 units.
3. Same Quadrant (same signs, like -12 and -4): Subtract the absolute values: |-12| - |-4| = 12 - 4 = 8 units.

Block Connection (COUNT-01): Cross-axis: 6 + 9 = 15 units. Same quadrant: 12 - 4 = 8 units. On SC-6, [subtract] then [g] [|x|] works for BOTH!

SC-6 Keystrokes: `6` `+/-` `ENTER` `9` `-` `g` `|x|`

Answer: 15 units; and 8 units

Subtracting and pressing [g] [|x|] works whether point..

SECTION 6.5 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Finding Horizontal & Vertical Distances on the Coordinate Plane..

1 Big Ideas Method: Cross-Axis / Same-Quadrant Distance

Find the horizontal distance between (-8, 5) and (11, 5).

Work:

Answer: _____

2 Big Ideas Method: Cross-Axis / Same-Quadrant Distance

Find the vertical distance between (-3, -4) and (-3, -15).

Work:

Answer: _____

3 Big Ideas Method: Cross-Axis / Same-Quadrant Distance

Find the vertical distance between (7, 9) and (7, -6).

Work:

Answer: _____

4 Big Ideas Method: Cross-Axis / Same-Quadrant Distance

Find the horizontal distance between (-14, 2) and (-5, 2).

Work:

Answer: _____

5 Big Ideas Method: Decimal & Fraction Coordinates

Find the distance between (-3.5, 4) and (6.5, 4).

Work:

Answer: _____

6 Big Ideas Method: Decimal & Fraction Coordinates

Find the distance between (2, -4 1/2) and (2, 3 1/2).

Work:

Answer: _____

7 Big Ideas Method: Decimal & Fraction Coordinates

A line segment starts at (-6, 3) and goes 10 units right. What are the coordinates of the right endpoint?

Work:

Answer: _____

8 Big Ideas Method: Decimal & Fraction Coordinates

Spot the Glitch: For (-5, 2) to (8, 2), Sam subtracted 8 - 5 = 3. Why is the distance 13, not 3?

Work:

Answer: _____

9 Application & 2-Way Check: Coordinate Polygon Perimeter/Area

A treasure map rectangle has corners (-4, 6), (5, 6), (5, -2), (-4, -2). Find its perimeter & area 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Coordinate Polygon Perimeter/Area

On a city grid (1 unit = 0.5 mile), you walk from (-6, -2) to (-6, 8) to (4, 8). How many miles did you walk?

Work:

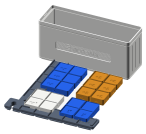
Answer: _____

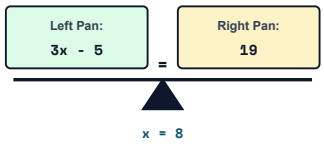
Section 7.1: Writing Equations & Testing Solutions — The Balance Scale Rule

Big Ideas Method: Writing Equations in One Variable & Testing Solutions by Substitution (Balance Scale Rule) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
In algebra, '=' is the center pivot of a Balance Scale! On your SC-6, Floor 2 (Y) is the Left Pan and Floor 1 (X) is the Right Pan. To test if x = 8 solves 3x - 5 = 19, put 19 on Y, compute 3(8) - 5 on X, and see Y: 19 | X: 19 (or press [g] [≠] -> 0 difference)!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & STAGE)





HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & STAGE):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and set the Balance Scale pivot (=) between the Left Pan and Right Pan.

Step 2: Place 4 x-rods and 5 unit cubes on the Left Pan (4x + 5) and 41 unit cubes on the Right Pan.

Step 3: Substitute x = 9 into each x-rod: 4(9) + 5 = 36 + 5 = 41, matching the Right Pan (41 = 41, Balanced!).

WORKED EXAMPLE — EXAMPLE 1: BALANCE SCALE SOLUTION TESTER

Problem: Test whether x = 7 or x = 9 is the solution to 4x + 5 = 41.

Big Ideas Textbook Method (Writing Equations in One Variable & Testing Solutions by Substitution (B..

1. Substitute the test value (x = 9) into the equation 4x + 5 = 41: 4(9) + 5 = 41.
2. Evaluate the left side using Order of Operations: 36 + 5 = 41.
3. Because 41 = 41 is a true statement, x = 9 is a solution.

Block Connection (ALGB-01): Park 41 on Y. Test x = 7: 4(7)+5 = 33 (Not 41!). Test x = 9: 4(9)+5 = 41 (Matches Y = 41—Balanced!).

SC-6 Keystrokes: `41` `ENTER` `4` `ENTER` `9` `×` `5` `+`

Answer: x = 9 is the solution (41 == *When Floor 2 (Y) and Floor 1 (X) match, the equation..*

SECTION 7.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Writing Equations in One Variable & Testing Solutions by Substi..

1 Big Ideas Method: Substitution Solution Test
Test whether x = 6 or x = 8 solves 5x - 7 = 33.

Work:

Answer: _____

2 Big Ideas Method: Substitution Solution Test
Test whether y = 12 or y = 15 solves y / 3 + 4 = 9.

Work:

Answer: _____

3 Big Ideas Method: Substitution Solution Test
Translate into an equation: 'The sum of a number x and 14 is 52.'

Work:

Answer: _____

4 Big Ideas Method: Substitution Solution Test
Translate into an equation: '6 times a number k equals 84.'

Work:

Answer: _____

5 Big Ideas Method: Translating Verbal Equations
Translate: '9 less than a number w is 17.' (Watch the order!)

Work:

Answer: _____

6 Big Ideas Method: Translating Verbal Equations
Translate: 'The quotient of a number m and 5 is 13.'

Work:

Answer: _____

7 Big Ideas Method: Translating Verbal Equations
4 pizza pies (p) plus a $6 delivery fee total $54. Write the equation and test p = 12!

Work:

Answer: _____

8 Big Ideas Method: Translating Verbal Equations
Spot the Glitch: For '12 less than x is 20', Sam wrote 12 - x = 20. Fix the equation and solve!

Work:

Answer: _____

9 Application & 2-Way Check: Real-World Equation Modeling
A streaming pass costs $10 plus $3 per movie (10 + 3m = 28). Find m by testing AND by undoing!

Work:

Answer: _____

10 Application & 2-Way Check: Real-World Equation Modeling
Three friends split a restaurant bill b equally and each pays $14.50. Write the equation and find b!

Work:

Answer: _____

BLOCK KIT: ALGB-01 • Algebra Boxes & Stage

Section 7.2: Solving One-Step Addition & Subtraction Equations ([f] [LASTX])

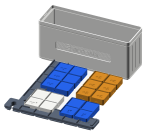
Big Ideas Method: Subtraction & Addition Properties of Equality (Inverse Operations on Both Sides)

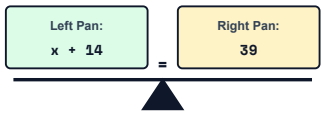
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Why does x + 14 = 39 become 39 - 14? Because the Subtraction Property of Equality removes 14 cubes from both pans of the balance scale so the x-rod stands alone! And why does x - 18 = 45 become 45 + 18? Because the Addition Property of Equality adds 18 back to both sides! Use [f] [LASTX] and the inverse operation to check!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & STAGE)





Remove 14 -> x = 25

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & STAGE):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and place 1 x-rod + 14 unit cubes on the Left Pan and 39 cubes on the Right Pan.

Step 2: Remove 14 unit cubes from BOTH pans to keep the scale balanced: 39 - 14 = 25 cubes remain.

Step 3: Read the isolated x-rod value: x = 25, and check by putting 14 back: 25 + 14 = 39.

WORKED EXAMPLE — EXAMPLE 1: UNDOING ADDITION (X + 28 = 73) + [F] [LASTX] PROOF

Problem: Solve x + 28 = 73 using the Subtraction Property of Equality and check your solution.

Big Ideas Textbook Method (Subtraction & Addition Properties of Equality (Inverse Operations on Bot..

1. Identify the operation applied to the variable in x + 28 = 73: + 28 (addition).
2. Apply the inverse operation (Subtraction Property of Equality) to both sides: x + 28 - 28 = 73 - 28 -> x = 45.
3. Check by substituting x = 45 into the original equation: 45 + 28 = 73.

Block Connection (ALGB-01): Remove 28 from both sides: 73 [ENTER] 28 [-] -> x = 45. Now tap [f] [LASTX] [+] -> 45 + 28 = 73!

SC-6 Keystrokes: 73 | ENTER | 28 | - | f | LASTX | +

Answer: x = 45 (Proven: 45 + 28 = 73) *Subtracting 28 from both sides isolates x = 45!*

SECTION 7.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Subtraction & Addition Properties of Equality (Inverse Operatio..

1 Big Ideas Method: Subtraction Property of Equality

Solve x + 28 = 73 and prove with [f] [LASTX] [+].

Work:

Answer: _____

2 Big Ideas Method: Subtraction Property of Equality

Solve y - 34 = 58 and prove with [f] [LASTX] [-].

Work:

Answer: _____

3 Big Ideas Method: Subtraction Property of Equality

Solve 19 + n = 64.

Work:

Answer: _____

4 Big Ideas Method: Subtraction Property of Equality

Solve k - 47 = 29.

Work:

Answer: _____

5 Big Ideas Method: Addition Property of Equality

Solve w - 19.6 = 34.8.

Work:

Answer: _____

6 Big Ideas Method: Addition Property of Equality

Solve p + 8.75 = 25.4.

Work:

Answer: _____

7 Big Ideas Method: Addition Property of Equality

Solve m + 3/8 = 5/6 on your SC-6 and check with [f] [LASTX] [+].

Work:

Answer: _____

8 Big Ideas Method: Addition Property of Equality

Solve a - 1 1/4 = 3 1/2.

Work:

Answer: _____

9 Application & 2-Way Check: 1-Step Inverse Equation & Check

After spending $18.50 on a controller skin, you have $26.75 left (m - 18.50 = 26.75). Find starting money m!

Work:

Answer: _____

10 Application & 2-Way Check: 1-Step Inverse Equation & Check

A drone climbs 42.5 ft to reach 110 ft (h + 42.5 = 110). Find its starting height h and prove with [f][LASTX]!

Work:

Answer: _____

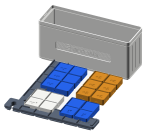
Section 7.3: Solving One-Step Multiplication & Division Equations

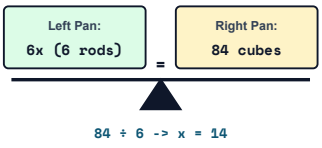
Big Ideas Method: Division & Multiplication Properties of Equality & Reciprocal Multiplication for Fractional Coefficients **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

When 6 identical x-rods balance 84 cubes (6x = 84), divide 84 into 6 equal piles (84 [ENTER] 6 [÷] -> 14)! When a variable is divided (k/7 = 14.5), multiply by 7 to put the 7 parts back together! And for a fraction coefficient like 2/5n = 16, either divide by 2/5 OR flip with [f] [1/x] and multiply!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA & FRACTION STUDIO)





HOW TO BUILD IT WITH ALGB-01 (ALGEBRA & FRACTION STUDIO):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and place 4 green x-rods on the Left Pan and 72 unit cubes on the Right Pan (4x = 72).

Step 2: Deal the 72 unit cubes equally across the 4 x-rods: 72 ÷ 4 = 18 cubes per x-rod.

Step 3: For (3/4)x = 18, three fourths of an x-rod equal 18 (so each fourth is 6, and the whole 4/4 x-rod is 4 × 6 = 24).

WORKED EXAMPLE — EXAMPLE 1: UNDOING MULTIPLICATION & DIVISION

Problem: Solve 8x = 136 and k / 7 = 14.5, and prove each with [f] [LASTX].

Big Ideas Textbook Method (Division & Multiplication Properties of Equality & Reciprocal Multiplicatio..

1. To solve 4x = 72, use the Division Property of Equality (divide both sides by 4): x = 72 ÷ 4 = 18.
2. To solve x / 6 = 14, use the Multiplication Property of Equality (multiply both sides by 6): x = 14 × 6 = 84.
3. To solve (3/4)x = 18, multiply both sides by the reciprocal 4/3: x = 18 × (4/3) = 24.

Block Connection (ALGB-01): For 8x = 136: 136 [ENTER] 8 [÷] -> 17 (Check: [f] [LASTX] [×] -> 136). For k/7 = 14.5: 14.5 [ENTER] 7 [×] -> 101.5.

SC-6 Keystrokes: `136` `ENTER` `8` `÷` `14.5` `ENTER` `7` `×`

Answer: x = 17; k = 101.5

Division undoes multiplication; multiplication undoes..

SECTION 7.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Division & Multiplication Properties of Equality & Reciprocal M..

1 Big Ideas Method: Division / Multiplication Property

Solve 8x = 136 and prove with [f] [LASTX] [×].

Work:

Answer: _____

2 Big Ideas Method: Division / Multiplication Property

Solve 12y = 180 and prove with [f] [LASTX] [×].

Work:

Answer: _____

3 Big Ideas Method: Division / Multiplication Property

Solve m / 6 = 19 and prove with [f] [LASTX] [÷].

Work:

Answer: _____

4 Big Ideas Method: Division / Multiplication Property

Solve k / 7 = 14.5.

Work:

Answer: _____

5 Big Ideas Method: Reciprocal Coefficient Solver

Solve 4.5w = 63.

Work:

Answer: _____

6 Big Ideas Method: Reciprocal Coefficient Solver

Solve 3/4 x = 18.

Work:

Answer: _____

7 Big Ideas Method: Reciprocal Coefficient Solver

Solve 2/5 n = 16 using BOTH [÷] and [f] [1/x] [×].

Work:

Answer: _____

8 Big Ideas Method: Reciprocal Coefficient Solver

Solve 5/6 p = 2/3.

Work:

Answer: _____

9 Application & 2-Way Check: Rate & Geometry Formula Equation

A 3D printer prints at 1.5 cm/min. How long (t) to print a 27 cm tower (1.5t = 27)? Solve & prove!

Work:

Answer: _____

10 Application & 2-Way Check: Rate & Geometry Formula Equation

3/5 of the students in a club (3/5 s = 27) are 6th graders. How many total students s are in the club?

Work:

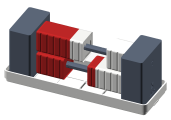
Answer: _____

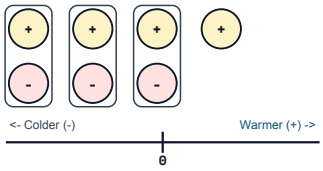
Section 7.4: Writing, Graphing & Solving Inequalities ([f] [>], [g] [<], [f] [≥], [g] [≤])

Big Ideas Method: Writing, Testing & Graphing Inequalities on a Number Line (Open Circle <, > vs. Closed Circle ≤, ≥) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Inequalities have infinitely many solutions! On a number line: '>' and '<' use an OPEN circle (o) because the boundary number is not included; '≥' (at least) and '≤' (at most) use a SOLID dot (•) because the boundary IS included! Solve just like one-step equations, then test boundary numbers with [f] [≥] ([f][+]) and [g] [≤] ([g][+])!

BLOCK KIT & STEP-BY-STEP GUIDE (RACK-01: NUMBER RACK & NUMBER LINE)





HOW TO BUILD IT WITH RACK-01 (NUMBER RACK & NUMBER LINE):

Step 1: Open Kit RACK-01 / ALGB-01 and mark the boundary value x = 5 on the number-line track.

Step 2: Use an OPEN ring for strict inequalities (< or >) and a SOLID cap for inclusive inequalities (≤ or ≥, 'at least' / 'at most').

Step 3: Test a value in the shaded direction (e.g., x = 8 for x ≥ 5) to confirm the inequality holds true.

WORKED EXAMPLE — EXAMPLE 1: SOLVING & GRAPHING X - 7 ≥ 12

Problem: Solve x - 7 ≥ 12, state whether the circle at 19 is Open or Solid, and test x = 20.

Big Ideas Textbook Method (Writing, Testing & Graphing Inequalities on a Number Line (Open Circle <,..

1. Translate key phrases: 'at least / no less than' -> ≥; 'at most / no more than' -> ≤.
2. Solve one-step inequalities using inverse operations just like equations: x + 7 ≤ 19 -> x ≤ 12.
3. Graph on a number line: Closed circle (●) for ≤ or ≥, Open circle (○) for < or >, and shade the true direction.

Block Connection (RACK-01): Add 7 to both sides: 12 [ENTER] 7 [+] -> x ≥ 19. Because it is '≥', put a SOLID dot (•) at 19 and shade RIGHT!

SC-6 Keystrokes: `12` `ENTER` `7` `+` `20` `x⇔y` `f` `≥`

Answer: x ≥ 19 (Solid dot at 19, arr.. *Testing 20 ≥ 19 with [f] [≥] outputs 1 (TRUE)!*

SECTION 7.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Writing, Testing & Graphing Inequalities on a Number Line (Open..

1 Big Ideas Method: Inequality Phrase & Solution Test
Solve x + 8 > 21. Tell whether the circle at 13 is Open or Solid, and arrow direction.

Work:

Answer: _____

2 Big Ideas Method: Inequality Phrase & Solution Test
Solve y - 9 ≤ 15. Tell whether the circle at 24 is Open or Solid, and arrow direction.

Work:

Answer: _____

3 Big Ideas Method: Inequality Phrase & Solution Test
Solve 6n ≥ 78. Tell whether the circle at 13 is Open or Solid, and arrow direction.

Work:

Answer: _____

4 Big Ideas Method: Inequality Phrase & Solution Test
Solve m / 4 < 9. Tell whether the circle at 36 is Open or Solid, and arrow direction.

Work:

Answer: _____

5 Big Ideas Method: Open (○) vs. Closed (●) Graph
Translate: 'You must be at least 48 inches tall (h) to ride.' Write the inequality!

Work:

Answer: _____

6 Big Ideas Method: Open (○) vs. Closed (●) Graph
Translate: 'An elevator holds at most 2,000 pounds (w).' Write the inequality!

Work:

Answer: _____

7 Big Ideas Method: Open (○) vs. Closed (●) Graph
Which numbers from {4, 5, 6} are solutions to 4n + 3 > 23? Test on your SC-6!

Work:

Answer: _____

8 Big Ideas Method: Open (○) vs. Closed (●) Graph
Which numbers from {4, 5, 6} are solutions to 4x - 3 ≥ 17?

Work:

Answer: _____

9 Application & 2-Way Check: 1-Step Inequality Modeling
You have $50 to buy $8 books (8b ≤ 50). Solve with [g] [÷R] to find the max whole books b you can buy!

Work:

Answer: _____

10 Application & 2-Way Check: 1-Step Inequality Modeling
A gamer needs at least 500 points (p + 340 ≥ 500). Solve for p AND test p = 160 with [f] [≥]!

Work:

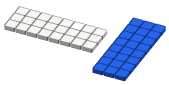
Answer: _____

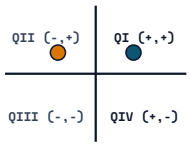
Section 7.5: Two-Variable Relationships (y = kx and y = x + b) — Tables & Graphs

Big Ideas Method: Independent (Input x) & Dependent (Output y) Variables, Function Tables & Rules (y = mx or y = x +.. Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
How do you tell if a table of (x, y) points is an ADDITION rule (y = x + b) or a MULTIPLICATION rule (y = kx)? Check y - x (if constant, it's y = x + b) and check y ÷ x (if constant, it's y = kx)! Remember: x is the Independent variable (input you choose) and y is the Dependent variable (output that depends on x).

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit ALGB-01 / COUNT-01 and build an Input-Output table where Input x = 1, 2, 3, 4.

Step 2: Compare how Output y changes: if y - x is constant (e.g., +7), the rule is additive (y = x + 7).

Step 3: If y ÷ x is constant (e.g., ×12), the rule is multiplicative (y = 12x).

WORKED EXAMPLE — EXAMPLE 1: THE 2-SECOND RULE DETECTOR (Y = KX VS. Y = X + B)

Problem: Table A has (2, 14), (5, 35), (8, 56). Table B has (10, 6), (15, 11), (22, 18). Find both equations.

Big Ideas Textbook Method (Independent (Input x) & Dependent (Output y) Variables, Function Tables..

1. Identify the Independent Variable x (the input you choose, such as hours) and Dependent Variable y (the output that depends on x).
2. Test the table pairs (x, y): if y - x is constant, write y = x + b; if y / x is constant, write y = mx.
3. Use the equation to find any missing output y or input x.

Block Connection (COUNT-01): Table A: 14÷2 = 7, 35÷5 = 7, 56÷8 = 7 -> Multiplicative y = 7x! Table B: 10 - 4 = 6, 15 - 4 = 11 -> Additive y = x - 4!

SC-6 Keystrokes: `14` `ENTER` `2` `÷` `35` `ENTER` `5` `÷`

Answer: Table A: y = 7x; Table B: y.. *Once you know y = 7x, find y at x = 12 with 7 [ENTE..*

SECTION 7.5 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Independent (Input x) & Dependent (Output y) Variables, Functio..

1 Big Ideas Method: Independent (x) vs. Dependent (y)
Table has (1, 8), (2, 16), (3, 24). Write the equation and find y when x = 10.

Work:

Answer: _____

2 Big Ideas Method: Independent (x) vs. Dependent (y)
Table has (1, 9), (2, 10), (5, 13). Write the equation and find y when x = 20.

Work:

Answer: _____

3 Big Ideas Method: Independent (x) vs. Dependent (y)
Table has (2, 14), (5, 35), (8, 56). Write the equation and find y when x = 12.

Work:

Answer: _____

4 Big Ideas Method: Independent (x) vs. Dependent (y)
Table has (10, 6), (15, 11), (22, 18). Write the equation and find y when x = 30.

Work:

Answer: _____

5 Big Ideas Method: Additive vs. Multiplicative Rule
Use 6 [g] [REP] [+] to generate the y-values of y = 6x for x = 1, 2, 3, 4.

Work:

Answer: _____

6 Big Ideas Method: Additive vs. Multiplicative Rule
In d = 65t (distance d in miles after t hours), which variable is Independent and which is Dependent?

Work:

Answer: _____

7 Big Ideas Method: Additive vs. Multiplicative Rule
Using d = 65t, how far does the car travel in 4.5 hours?

Work:

Answer: _____

8 Big Ideas Method: Additive vs. Multiplicative Rule
Does the graph of y = 4x pass through the origin (0, 0)? What about y = x + 4?

Work:

Answer: _____

9 Application & 2-Way Check: Table-to-Equation Prediction
You earn p = 12.50h. How many hours h must you work to earn $100? Solve by division AND ratio doubling!

Work:

Answer: _____

10 Application & 2-Way Check: Table-to-Equation Prediction
A gym charges a $25 signup fee plus $15 per month (m). Write the equation for total cost c and find c for 12 months!

Work:

Answer: _____

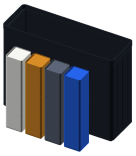
BLOCK KIT: COUNT-01 • 3x1 Gridfinity Box & Tiles

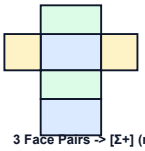
Section 8.1: 3D Solids, Nets & Surface Area of Rectangular Prisms ([f] [RECT] + [Σ+])

Big Ideas Method: Parallelogram Cut-and-Slide (A = bh) & Triangle as Half of a Parallelogram (A = ½bh) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Finding the surface area of a 6-sided box on paper is a magnet for mistakes because kids forget to double one of the 3 face pairs! Touch your physical Gridfinity box: clear memory ([g] [CLΣ]), compute each pair of opposite walls with l [ENTER] w [f] [RECT] 2 [×] [Σ+] (watch n=1, n=2, n=3!), and read the total with [f] [Σx]!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: 3X1 GRIDFINITY BOX & TILES)





HOW TO BUILD IT WITH COUNT-01 (3X1 GRIDFINITY BOX & TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and build a 12 × 8 rectangle array (96 unit squares).

Step 2: For a parallelogram, slide the right-triangle overhang from one side to the other—area is still Base × Perpendicular Height (12 × 8 = 96)!

Step 3: Cut the 12 × 8 rectangle diagonally into 2 congruent triangles—each triangle is exactly half: ½ × 12 × 8 = 48.

WORKED EXAMPLE — EXAMPLE 1: THE 3-PAIR BOX LEDGER ([F] [RECT] 2 [×] [Σ+])

Problem: Find the surface area of a rectangular prism measuring 8 cm by 5 cm by 3 cm.

Big Ideas Textbook Method (Parallelogram Cut-and-Slide (A = bh) & Triangle as Half of a Parallelogram..

1. Parallelogram Area: Multiply the base b by the perpendicular height h (ignore the slanted side!): A = b × h = 12 × 8 = 96.
2. Triangle Area: Take half of the base × height rectangle: A = ½ × b × h.
3. Missing Height/Base: For a triangle, double the area first (2A), then divide by the known base or height.

Block Connection (COUNT-01): Clear ([g] [CLΣ]). Top/Bot: 8×5×2 = 80 [Σ+] (n=1). Front/Back: 8×3×2 = 48 [Σ+] (n=2). Left/Right: 5×3×2 = 30 [Σ+] (n=3). Tap [f] [Σx]!

SC-6 Keystrokes:

`g` `CLΣ` `8` `ENTER` `5` `f` `RECT` `2` `×` `Σ+` `8` `ENTER` `3` `f` `RECT` `2` `×` `Σ+` `5` `ENTER` `3` `f` `RECT` `2` `×` `Σ+` `f` `Σx`

Answer: 158 cm²

Seeing n = 3 guarantees you logged all 3 pairs of fac..

SECTION 8.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Parallelogram Cut-and-Slide (A = bh) & Triangle as Half of a Pa..

1 Big Ideas Method: Parallelogram Area (A = bh)
Find the surface area of a cube with edge length s = 4 cm (6s²).

Work:

Answer: _____

2 Big Ideas Method: Parallelogram Area (A = bh)
Find the surface area of a cube with edge length s = 9 in.

Work:

Answer: _____

3 Big Ideas Method: Parallelogram Area (A = bh)
Find the surface area of a 6 cm by 4 cm by 3 cm rectangular prism using the 3-pair [Σ+] ledger.

Work:

Answer: _____

4 Big Ideas Method: Parallelogram Area (A = bh)
Find the surface area of a 10 in by 5 in by 2 in rectangular box.

Work:

Answer: _____

5 Big Ideas Method: Triangle Area (A = ½bh)
Find the surface area of a prism with l = 12 ft, w = 8 ft, and h = 5 ft.

Work:

Answer: _____

6 Big Ideas Method: Triangle Area (A = ½bh)
Find the surface area of a prism with decimal edges: 5 m by 4 m by 2.5 m.

Work:

Answer: _____

7 Big Ideas Method: Triangle Area (A = ½bh)
A cube has a total surface area of 150 cm². What is the area of 1 face, and what is the edge length s?

Work:

Answer: _____

8 Big Ideas Method: Triangle Area (A = ½bh)
Spot the Glitch: For a 5×4×3 box, Leo added 20 + 15 + 12 = 47. What did he forget to do?

Work:

Answer: _____

9 Application & 2-Way Check: Missing Height & Shaded Area
An open-top 3D-printed bin is 10 cm long, 8 cm wide, and 5 cm tall (NO lid!). Find its outer surface area 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Missing Height & Shaded Area
You wrap a gift box (15 in × 10 in × 4 in). If you have 550 in² of wrapping paper, is it enough?

Work:

Answer: _____

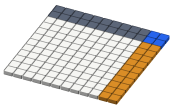
Section 8.2: Surface Area of Triangular Prisms & Pyramids ([f] [RECT] + [g] [TRI] + [Σ+])

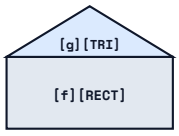
Big Ideas Method: Trapezoid Area Formula A = ½h(b1 + b2) & Decomposing Composite Figures into Rectangles/Triang.. Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Never memorize complicated formulas for triangular prisms or pyramids! Just look at the flat shapes in the unfolded net: use [f] [RECT] for rectangular walls and [g] [TRI] for triangular faces, tossing each piece into [Σ+] and reading the sum with [f] [Σx]!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





Sum chunks with [Σ+]!

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and place two identical trapezoids (bases b1 = 6, b2 = 10, height h = 5) together.

Step 2: Flip the second trapezoid upside-down beside the first to form a parallelogram of base (6 + 10 = 16) and height 5 (Area = 80).

Step 3: Take half of that parallelogram for 1 trapezoid: ½ × (6 + 10) × 5 = 40! For L-shapes, split into non-overlapping rectangles.

WORKED EXAMPLE — EXAMPLE 1: SQUARE PYRAMID TENT (1 SQUARE + 4 TRIANGLES)

Problem: Find the surface area of a square pyramid with a 10 m × 10 m base and 4 triangles (b = 10 m, slant height = 12 m).

Big Ideas Textbook Method (Trapezoid Area Formula A = ½h(b1 + b2) & Decomposing Composite Figu..

1. Trapezoid Area: Add the two parallel bases (b1 + b2), multiply by the height h, and divide by 2: A = ½ × 5 × (6 + 10) = 40.
2. Composite Figures: Draw a dashed line to split the shape into simple rectangles, triangles, or trapezoids.
3. Compute the area of each piece and add them together (or subtract a cutout from the outer rectangle).

Block Connection (COUNT-01): Clear ([g] [CLΣ]). Square Base: 10 [x²] [Σ+] (100). 4 Triangles: 10 [ENTER] 12 [g] [TRI] 4 [×] [Σ+] (240). Read [f] [Σx] -> 340 m²!

SC-6 Keystrokes: g CLΣ 10 x² Σ+ 10 ENTER 12 g TRI 4 × Σ+ f Σx

Answer: 340 m² *1 Square Base (100) + 4 Identical Triangles (4 × 60 =..*

SECTION 8.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Trapezoid Area Formula A = ½h(b1 + b2) & Decomposing Composite..

1 Big Ideas Method: Trapezoid Average-Base Formula

Square Pyramid: Base is 6 cm × 6 cm; 4 triangles each have b = 6 cm, slant height = 5 cm. Find SA.

Work:

Answer: _____

2 Big Ideas Method: Trapezoid Average-Base Formula

Square Pyramid: Base is 8 in × 8 in; 4 triangles each have b = 8 in, slant height = 10 in. Find SA.

Work:

Answer: _____

3 Big Ideas Method: Trapezoid Average-Base Formula

Square Pyramid: Base is 12 ft × 12 ft; 4 triangles each have b = 12 ft, slant height = 9 ft. Find SA.

Work:

Answer: _____

4 Big Ideas Method: Trapezoid Average-Base Formula

Regular Tetrahedron: Made of 4 identical equilateral triangles (b = 6 cm, h = 5.2 cm). Find total SA.

Work:

Answer: _____

5 Big Ideas Method: Composite Split / Cutout Area

Triangular Prism: 2 triangle ends (b=4, h=3) + 3 rectangles of length 10 (widths 3, 4, 5). Find SA.

Work:

Answer: _____

6 Big Ideas Method: Composite Split / Cutout Area

Triangular Prism Tent: 2 triangle ends (b=8 ft, h=3 ft) + 3 rectangles of length 7 ft (widths 5, 5, 8 ft).

Work:

Answer: _____

7 Big Ideas Method: Composite Split / Cutout Area

A regular tetrahedron has 4 identical triangles with b = 8 cm and h = 6.93 cm. Find its surface area.

Work:

Answer: _____

8 Big Ideas Method: Composite Split / Cutout Area

How many faces (triangles and rectangles) does a Triangular Prism have vs. a Square Pyramid?

Work:

Answer: _____

9 Application & 2-Way Check: 2-Way Decomposition Proof

Camping Tent (Triangular Prism): 2 ends (b=6 ft, h=4 ft), floor (6×8 ft), and 2 roof sides (5×8 ft each). Find total fabric 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: 2-Way Decomposition Proof

A glass pyramid skylight has NO bottom floor—only 4 triangular glass panes (b = 9 ft, slant h = 6 ft). Find glass area!

Work:

Answer: _____

Section 8.3: Volume of Rectangular Prisms with Whole & Fractional Edges ([f] [PRISM])

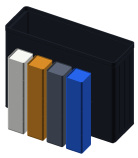
Big Ideas Method: Unfolding 3D Solids into 2D Nets — Rectangular Prisms SA = 2(lw + lh + wh) & Pyramids

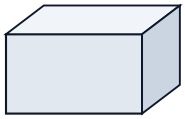
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Surface Area is the flat cardboard on the outside; Volume is how many 3D unit cubes fit inside! On key 9 of your SC-6, [f] [PRISM] ([f][9]) takes Length (l) on Floor 3 (Z), Width (w) on Floor 2 (Y), and Height (h) on Floor 1 (X) and multiplies all three dimensions in a single click!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & FRACTION STUDIO)





l [ENTER] w [ENTER] h [f][PRISM]

HOW TO BUILD IT WITH COUNT-01 (COUNTING & FRACTION STUDIO):

Step 1: Open Kit COUNT-01 / ALGB-01 and inspect a 3D Gridfinity box with length l = 8, width w = 5, and height h = 4.

Step 2: Unfold its 6 faces flat into 3 congruent opposite pairs: Bottom/Top (8×5 = 40 each), Front/Back (8×4 = 32 each), Left/Right (5×4 = 20 each).

Step 3: Add the 3 pairs: 2(40) + 2(32) + 2(20) = 80 + 64 + 40 = 184 sq units.

WORKED EXAMPLE — EXAMPLE 1: 1-CLICK 3-FLOOR VOLUME WITH [F] [PRISM]

Problem: Find the volume of a shipping crate measuring 12 in × 8 in × 5 in, and an aquarium 3 1/2 ft × 1 1/2 ft × 2 ft.

Big Ideas Textbook Method (Unfolding 3D Solids into 2D Nets — Rectangular Prisms SA = 2(lw + lh +..

1. Rectangular Prism Net (6 faces): Find the 3 face products lw = 40, lh = 32, wh = 20, sum them (92), and double: SA = 2(92) = 184.
2. Cube Net (6 identical squares): SA = 6s².
3. Square Pyramid Net (1 square base + 4 identical triangular faces): SA = b² + 4(½ × b × l).

Block Connection (COUNT-01): Push l [ENTER] w [ENTER] h onto Z, Y, X and press [f] [PRISM] ([f][9])!

SC-6 Keystrokes: 12 | ENTER | 8 | ENTER | 5 | f | PRISM

Answer: Crate = 480 in³; Aquarium =. *For the aquarium: 3.5 [ENTER] 1.5 [ENTER] 2 [f] [PR..*

SECTION 8.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Unfolding 3D Solids into 2D Nets — Rectangular Prisms SA = 2(lw..

1 Big Ideas Method: 3-Pair Prism Net SA = 2(lw+lh+wh)

Use [f] [PRISM] to find the volume of a box measuring 10 cm by 6 cm by 4 cm.

Work:

Answer: _____

2 Big Ideas Method: 3-Pair Prism Net SA = 2(lw+lh+wh)

Find the volume of a cube with edge length 5 in using [g] [x³] or [f] [PRISM].

Work:

Answer: _____

3 Big Ideas Method: 3-Pair Prism Net SA = 2(lw+lh+wh)

Find the volume of a crate measuring 15 ft by 8 ft by 6 ft.

Work:

Answer: _____

4 Big Ideas Method: 3-Pair Prism Net SA = 2(lw+lh+wh)

A prism has Base Area B = 32 m² and height h = 7 m. Find its volume.

Work:

Answer: _____

5 Big Ideas Method: Cube & Pyramid Net Surface Area

Fractional Edges: Find the volume of a fish tank 3 1/2 ft long, 1 1/2 ft wide, and 2 ft tall.

Work:

Answer: _____

6 Big Ideas Method: Cube & Pyramid Net Surface Area

Fractional Edges: Find the volume of a block 4 1/2 in by 3 in by 2 in.

Work:

Answer: _____

7 Big Ideas Method: Cube & Pyramid Net Surface Area

Missing Height: A box has Volume = 270 in³, length = 9 in, and width = 5 in. Find height h.

Work:

Answer: _____

8 Big Ideas Method: Cube & Pyramid Net Surface Area

How many 1/2-inch mini-cubes fit inside 1 cubic inch? (Compute 2³!)

Work:

Answer: _____

9 Application & 2-Way Check: Open-Top Box / Wrapping Area

A planter has Base Area 45 in² (9×5) and Volume 360 in³. Find depth h by division AND verify with [f] [PRISM]!

Work:

Answer: _____

10 Application & 2-Way Check: Open-Top Box / Wrapping Area

How many 1/2-inch cubes fit inside a box with volume 27 in³? (Multiply 27 by 2³ = 8!)

Work:

Answer: _____

Section 8.4: Circles, Cones & Right-Triangle Hypotenuse ([π], [f] [CIRC], [g] [CONE], [g] [HYP])

Big Ideas Method: 3D Layer Stacking Volume (V = B · h = l · w · h) & Fractional Unit-Cube Packing

Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Look at keys 8 and 9 on your SC-6: they hold 1-click geometry superpowers! For a circle of radius r, press r [f] [CIRC] ([f][8]) to compute πr² instantly. For a right triangle with legs a and b, press a [ENTER] b [g] [HYP] ([g][8]) for √(a²+b²). For a cone with radius r and height h, press r [ENTER] h [g] [CONE] ([g][9]) for ⅓πr²h!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





[g] [TRI] = ½ × b × h

HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 / FRAC-01 and cover the bottom of a 5 × 4 prism with 1 flat layer of 5 × 4 = 20 unit cubes (Base Area B = 20).

Step 2: Stack 3 identical layers high (h = 3) to count the total volume: V = 20 × 3 = 60 cubic units.

Step 3: For 1/2-inch cubes, 2 cubes fit along every 1 inch of length, width, and height (2 × 2 × 2 = 8 small cubes per 1 in³).

WORKED EXAMPLE — EXAMPLE 1: CIRCLE AREA ([F] [CIRC]) & 3-4-5 HYPOTENUSE ([G]..

Problem: Find the area of a pizza with radius r = 6 in, and the diagonal of a 12 ft by 5 ft ramp.

Big Ideas Textbook Method (3D Layer Stacking Volume (V = B · h = l · w · h) & Fractional Unit-Cube Pa..

1. Find the area of the base layer: B = l × w, then multiply by the height h: V = l × w × h.
2. Works identically when edges are fractions or decimals: multiply l × w × h.
3. To find a missing height h when given Volume V, divide V by the Base Area (l × w): h = V ÷ (l × w).

Block Connection (COUNT-01): For r = 6: press 6 [f] [CIRC] -> 113.10 in². For legs 12 and 5: press 12 [ENTER] 5 [g] [HYP] -> 13 ft!

SC-6 Keystrokes: 6 | f | CIRC | 12 | ENTER | 5 | g | HYP

Answer: Pizza = 113.10 in²; Ramp dia. *Check 6 [f] [CIRC] against 6 [x²] [π] [×] -> identical 11..*

SECTION 8.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: 3D Layer Stacking Volume (V = B · h = l · w · h) & Fractional U..

1 Big Ideas Method: Base-Layer Volume (V = lwh)

Find the area of a circle with radius r = 5 cm using [f] [CIRC].

Work:

Answer: _____

2 Big Ideas Method: Base-Layer Volume (V = lwh)

Find the area of a circle with radius r = 10 in using [f] [CIRC].

Work:

Answer: _____

3 Big Ideas Method: Base-Layer Volume (V = lwh)

Find the hypotenuse of a right triangle with legs a = 3 and b = 4 using [g] [HYP].

Work:

Answer: _____

4 Big Ideas Method: Base-Layer Volume (V = lwh)

Find the hypotenuse of a right triangle with legs a = 6 ft and b = 8 ft using [g] [HYP].

Work:

Answer: _____

5 Big Ideas Method: Fractional Edge & Cube Packing

Find the hypotenuse of a skate ramp with base 12 ft and back height 5 ft using [g] [HYP].

Work:

Answer: _____

6 Big Ideas Method: Fractional Edge & Cube Packing

A circle has diameter d = 16 m. Divide by 2 to find radius r, then press [f] [CIRC]!

Work:

Answer: _____

7 Big Ideas Method: Fractional Edge & Cube Packing

Find the volume of a cone with radius r = 3 cm and height h = 12 cm using [g] [CONE].

Work:

Answer: _____

8 Big Ideas Method: Fractional Edge & Cube Packing

Find the volume of a cone with radius r = 6 in and height h = 5 in using [g] [CONE].

Work:

Answer: _____

9 Application & 2-Way Check: Missing Dimension (h = V ÷ B)

Find the volume of a cone (r = 3 cm, h = 10 cm) both with [g] [CONE] AND via 3 [f] [CIRC] 10 [×] 3 [÷]!

Work:

Answer: _____

10 Application & 2-Way Check: Missing Dimension (h = V ÷ B)

TV screens are measured along the diagonal. A screen is 32 in wide and 24 in tall. Find its diagonal size 2 ways!

Work:

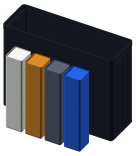
Answer: _____

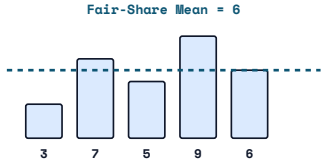
Section 9.1: Dot Plots, Frequency & Mean as 'Fair Share Leveling' ([Σ+], [g] [x-bar])

Big Ideas Method: Statistical Questions (Variability), Dot Plots & Fair-Share Leveling for the Mean (x-bar = Σx / n) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
What is the Mean (average) physically? If you build 5 unequal cube towers (3, 7, 5, 9, 6) and move cubes from the tallest towers onto the shortest ones until all 5 are the exact same height, every tower stands 6 cubes tall! On the SC-6, clear stats ([g] [CLΣ]), enter each number with [Σ+], and press [g] [x-bar] ([g][N])!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & ARRAY TILES)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & ARRAY TILES):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and build 5 vertical tile towers of heights 3, 7, 5, 9, and 6.

Step 2: Move top tiles from the taller towers (9 and 7) onto the shorter towers (3 and 5) until all 5 towers stand at the exact same height.

Step 3: All 5 leveled towers stand 6 tiles tall—proving the Fair-Share Mean is (3+7+5+9+6) ÷ 5 = 30 ÷ 5 = 6!

WORKED EXAMPLE — EXAMPLE 1: FAIR-SHARE MEAN WITH [Σ+] & [G] [x-bar]

Problem: Find the sum ([f] [Σx]), count ([N]), and Mean ([g] [x-bar]) of 18, 24, 15, 27, 21.

Big Ideas Textbook Method (Statistical Questions (Variability), Dot Plots & Fair-Share Leveling for the..

1. Sum all data values in the set: Σx = 3 + 7 + 5 + 9 + 6 = 30.
2. Count the number of data values: n = 5.
3. Divide the sum by the count: Mean x-bar = 30 ÷ 5 = 6.

Block Connection (COUNT-01): Press [g] [CLΣ], then enter each score followed by [Σ+]. Tap [f] [Σx] for total (105), [N] for count (5), and [g] [x-bar] for Mean (21)!

SC-6 Keystrokes: g | CLΣ | 18 | Σ+ | 24 | Σ+ | 15 | Σ+ | 27 | Σ+ | 21 | Σ+ | g | x̄

Answer: Sum = 105, Count = 5, Mean = *If you make a typo (e.g. 35 instead of 25), type 35 [f] [..*

SECTION 9.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Statistical Questions (Variability), Dot Plots & Fair-Share Leveling for the Mean (x-bar = Σx / n)

1 Big Ideas Method: Fair-Share Mean (Σx ÷ n)
Level the towers 2, 4, 6, 8. Find the sum ([f] [Σx]) and Mean ([g] [x-bar]).

Work:

Answer: _____

2 Big Ideas Method: Fair-Share Mean (Σx ÷ n)
Find the sum and Mean of: 12, 15, 18, 11, 14.

Work:

Answer: _____

3 Big Ideas Method: Fair-Share Mean (Σx ÷ n)
Find the Mean of your 5 gaming scores: 18, 24, 15, 27, 21.

Work:

Answer: _____

4 Big Ideas Method: Fair-Share Mean (Σx ÷ n)
Find the Mean of these 4 decimal weights (g): 4.2, 3.8, 4.5, 3.5.

Work:

Answer: _____

5 Big Ideas Method: Dot Plot Peak, Cluster & Gap
Typo Rescue: Enter 10 [Σ+] 20 [Σ+] 35 [Σ+], then remove 35 with 35 [f] [Σ-] and add 25 [Σ+]. Find [g] [x-bar]!

Work:

Answer: _____

6 Big Ideas Method: Dot Plot Peak, Cluster & Gap
Frequency Shortcut ([FRQ]): If 10 appears 4 times and 20 appears 2 times, find the Mean of all 6 numbers!

Work:

Answer: _____

7 Big Ideas Method: Dot Plot Peak, Cluster & Gap
Find the Mean daily high temperature (°F): 68, 72, 75, 71, 64, 70.

Work:

Answer: _____

8 Big Ideas Method: Dot Plot Peak, Cluster & Gap
If 5 towers have a Mean height of 12 blocks, how many total blocks are in all 5 towers combined?

Work:

Answer: _____

9 Application & 2-Way Check: Missing Target Score via Σx
You need a Mean of 90 across 4 quizzes (360 pts). Your first 3 are 85, 92, 88. Find Quiz 4 score 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Missing Target Score via Σx
6 players have a Mean score of 15 pts (90 pts). A 7th player scores 29 pts. What is the new Mean of all 7?

Work:

Answer: _____

Section 9.2: Median, Mode & Range — One Data Entry, Five Instant Answers!

Big Ideas Method: Ordering Data Least-to-Greatest for Median (Middle), Mode (Most Frequent) & Outlier Impact

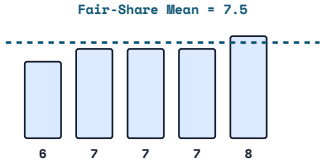
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Normally, finding Median and Range requires sorting the list by hand first—and missing one number ruins the answer! On the SC-6, enter your numbers with [Σ+] in ANY order: the chip sorts them automatically and gives you Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), and Range ([RNG]) in 1 click each!

BLOCK KIT & STEP-BY-STEP GUIDE (PROB-01: PROBABILITY & STATS STUDIO)





HOW TO BUILD IT WITH PROB-01 (PROBABILITY & STATS STUDIO):

Step 1: Open Kit COUNT-01 / PRIME-01 and line up the data towers in strict height order from shortest to tallest.

Step 2: Slide one finger in from the left end and one from the right end simultaneously until they meet at the middle tower (Median).

Step 3: If two middle towers remain (even n), average their heights; notice that making the tallest tower huge (an outlier) barely moves the Median!

WORKED EXAMPLE — EXAMPLE 1: ONE ENTRY, FIVE INSTANT STATS

Problem: For shoe sizes 7, 9, 6, 7, 10, 8, 9, 7, find Mean, Median, Mode, Min, Max, and Range.

Big Ideas Textbook Method (Ordering Data Least-to-Greatest for Median (Middle), Mode (Most Frequen..

1. ALWAYS order the data from least to greatest first.
2. Median: Middle value (if n is odd) or mean of the two middle values (if n is even). Mode: Value(s) that occur most often.
3. Best Measure of Center: Use the Median when an extreme outlier skews the Mean; use the Mean when data has no outlier.

Block Connection (PROB-01): Clear ([g] [CLΣ]), enter all 8 numbers with [Σ+], then tap [g] [x-bar], [f] [MED], [g] [MODE], and [RNG]!

SC-6 Keystrokes:

g	CLΣ	7	Σ+	9	Σ+	6	Σ+	7	Σ+	10	Σ+
8	Σ+	9	Σ+	7	Σ+	f	MED	g	MODE	RNG	

Answer: Mean = 7.875, Median = 7.5,.. *You don't even have to sort the numbers before typin..*

SECTION 9.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Ordering Data Least-to-Greatest for Median (Middle), Mode (Most..

1 Big Ideas Method: Ordered Median (Odd/Even n) & Mode

Load 4, 9, 2, 7, 8 into [Σ+]. Find the Median ([f] [MED]) and Range ([RNG]).

Work:

Answer: _____

2 Big Ideas Method: Ordered Median (Odd/Even n) & Mode

Load 12, 15, 12, 18, 20, 13 into [Σ+]. Find the Median, Mode ([g] [MODE]), and Range.

Work:

Answer: _____

3 Big Ideas Method: Ordered Median (Odd/Even n) & Mode

Find the Mean, Median, Mode, and Range of: 5, 8, 8, 10, 14.

Work:

Answer: _____

4 Big Ideas Method: Ordered Median (Odd/Even n) & Mode

Find the Minimum ([f] [MIN]), Maximum ([g] [MAX]), and Range ([RNG]) of: 23, 45, 19, 38, 51.

Work:

Answer: _____

5 Big Ideas Method: Outlier Impact on Mean vs. Median

Even Count Median: Find the Median of the 6 test scores 70, 80, 84, 90, 92, 98.

Work:

Answer: _____

6 Big Ideas Method: Outlier Impact on Mean vs. Median

Outlier Check: Earnings are $15, $18, $15, $20, $17, $90. Find Mean and Median. Which represents a typical day?

Work:

Answer: _____

7 Big Ideas Method: Outlier Impact on Mean vs. Median

What happens to the Mean and Median of {10, 12, 14, 16, 18} if 18 changes to 100?

Work:

Answer: _____

8 Big Ideas Method: Outlier Impact on Mean vs. Median

Mystery Range: The minimum score on a quiz is 64 and the Range is 31. What was the maximum score?

Work:

Answer: _____

9 Application & 2-Way Check: Choosing Best Measure of Center

Find Median and Range of {14, 8, 21, 11, 16} both by sorting tiles by hand AND with [f] [MED] & [RNG]!

Work:

Answer: _____

10 Application & 2-Way Check: Choosing Best Measure of Center

Home prices on a block (in $1,000s): 210, 220, 215, 225, 980. Find Mean and Median—which should a buyer trust?

Work:

Answer: _____

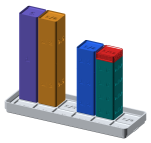
Section 9.3: Quartiles, Interquartile Range ([IQR]) & Box-and-Whisker Plots

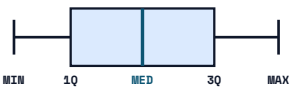
Big Ideas Method: Measures of Variation — Range (Max - Min), Quartiles (Q1, Q3) & Interquartile Range (IQR = Q3 - Q1) Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

A Box-and-Whisker Plot splits your sorted data into 4 equal quarters (25% in each section, just like four 1/4 Fraction Studio blocks!). Once your data is in [Σ+], tap [f] [MIN], [f] [1Q], [f] [MED], [g] [3Q], [g] [MAX], and [IQR] (= 3Q - 1Q) to get the entire Box Plot in 5 seconds!

BLOCK KIT & STEP-BY-STEP GUIDE (FRAC-01: FRACTION & STATS STUDIO)





Box = Middle 50% ([IQR])

HOW TO BUILD IT WITH FRAC-01 (FRACTION & STATS STUDIO):

Step 1: Open Kit COUNT-01 and arrange 7 ordered data tiles from Min to Max.

Step 2: Split the row at the center Median tile, then find the middle of the lower half (1st Quartile Q1) and upper half (3rd Quartile Q3).

Step 3: Measure the total spread (Range = Max - Min) and the middle-50% spread (IQR = Q3 - Q1).

WORKED EXAMPLE — EXAMPLE 1: INSTANT 5-NUMBER SUMMARY & [IQR]

Problem: Find the 5-Number Summary and IQR of: 68, 72, 75, 78, 82, 85, 88, 92, 96.

Big Ideas Textbook Method (Measures of Variation — Range (Max - Min), Quartiles (Q1, Q3) & Interquar..

1. Order the data from least to greatest and find the Minimum, Maximum, and Median (Q2).
2. Find Q1 (median of the lower half) and Q3 (median of the upper half).
3. Compute Range = Max - Min and Interquartile Range IQR = Q3 - Q1 (spread of the middle 50% of data).

Block Connection (FRAC-01): Load all 9 scores with [Σ+]. Read off [f] [MIN] (68), [f] [1Q] (73.5), [f] [MED] (82), [g] [3Q] (90), [g] [MAX] (96), and [IQR] (16.5)!

SC-6 Keystrokes: f MIN f 1Q f MED g 3Q g MAX IQR

Answer: Min=68, Q1=73.5, Med=82, Q3= *Check: [g] [3Q] (90) minus [f] [1Q] (73.5) = 16.5 ([IQR..*

SECTION 9.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Measures of Variation — Range (Max - Min), Quartiles (Q1, Q3) &..

1 Big Ideas Method: Range & 5-Number Summary

Load 10, 20, 30, 40, 50 into [Σ+]. Record Min, Q1, Med, Q3, Max, and [IQR].

Work:

Answer: _____

2 Big Ideas Method: Range & 5-Number Summary

Load 4, 6, 8, 10, 12, 14, 16 into [Σ+]. Find Q1, Median, Q3, and [IQR].

Work:

Answer: _____

3 Big Ideas Method: Range & 5-Number Summary

If a box plot has Q1 = 24 and Q3 = 58, what is the Interquartile Range (IQR)?

Work:

Answer: _____

4 Big Ideas Method: Range & 5-Number Summary

What percent of data in a Box Plot falls above Q1 (the 25th percentile)?

Work:

Answer: _____

5 Big Ideas Method: Quartiles (Q1, Q3) & IQR

Find the 5-Number Summary and [IQR] of the 9 scores: 68, 72, 75, 78, 82, 85, 88, 92, 96.

Work:

Answer: _____

6 Big Ideas Method: Quartiles (Q1, Q3) & IQR

Find Q1, Median, Q3, and [IQR] for the 8 speeds: 42, 45, 48, 50, 52, 55, 58, 60.

Work:

Answer: _____

7 Big Ideas Method: Quartiles (Q1, Q3) & IQR

Consistency Check: Player A has IQR = 4 pts and Player B has IQR = 19 pts. Who is more consistent?

Work:

Answer: _____

8 Big Ideas Method: Quartiles (Q1, Q3) & IQR

Outlier Rule (1.5 × IQR): If Q1 = 40, Q3 = 60 (IQR = 20), any score above Q3 + 1.5(IQR) is an outlier. Find that cutoff!

Work:

Answer: _____

9 Application & 2-Way Check: Consistency & Spread Comparison

For {12, 15, 18, 22, 25, 28, 34}, find [IQR] both with the [IQR] key AND by subtracting [g] [3Q] - [f] [1Q]!

Work:

Answer: _____

10 Application & 2-Way Check: Consistency & Spread Comparison

200 students take a test. The box plot has Min=50, Q1=70, Med=80, Q3=90, Max=100. How many students scored between 70 and 90?

Work:

Answer: _____

Section 9.4: Mean Absolute Deviation ([g] [MAD]) Without the 20-Minute Nightmare

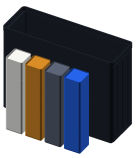
Big Ideas Method: Mean Absolute Deviation (MAD) — Average Distance of Data Values from the Mean

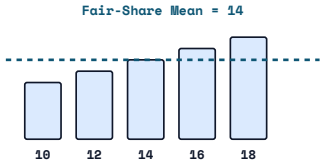
Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Every 6th grader dreads Mean Absolute Deviation (MAD) because doing it by hand takes 15 steps: find the Mean, subtract the Mean from every number, take the absolute value (|x|), and average all the distances! On the SC-6, once your data is in [Σ+], pressing [g] [MAD] ([g][Σ+]) computes the exact MAD in 1 click!

BLOCK KIT & STEP-BY-STEP GUIDE (COUNT-01: COUNTING & STATS STUDIO)





HOW TO BUILD IT WITH COUNT-01 (COUNTING & STATS STUDIO):

Step 1: Open Kit COUNT-01 (Counting & Array Tiles) and build towers for 4, 6, 8, 10, 12 alongside the Mean line at x-bar = 8.

Step 2: Count how many tiles each tower sits above or below the Mean line (distances |x - 8|): 4, 2, 0, 2, and 4 tiles.

Step 3: Find the average of those 5 distances: (4 + 2 + 0 + 2 + 4) ÷ 5 = 12 ÷ 5 = 2.4 tiles!

WORKED EXAMPLE — EXAMPLE 1: UNDERSTANDING & CHECKING MAD WITH [G] [MAD]

Problem: Find the Mean and Mean Absolute Deviation (MAD) of 10, 12, 14, 16, 18.

Big Ideas Textbook Method (Mean Absolute Deviation (MAD) — Average Distance of Data Values from..

1. Step 1 (Mean): Find the mean of the data set: (4 + 6 + 8 + 10 + 12) ÷ 5 = 8.
2. Step 2 (Absolute Deviations): Find the distance of each value from 8: |4-8|=4, |6-8|=2, |8-8|=0, |10-8|=2, |12-8|=4.
3. Step 3 (Mean of Distances): Average the distances: (4 + 2 + 0 + 2 + 4) ÷ 5 = 12 ÷ 5 = 2.4.

Block Connection (COUNT-01): Mean is 14. The distances |x - 14| from the Mean are 4, 2, 0, 2, 4 (sum = 12). Average distance = 12 ÷ 5 = 2.4! Confirm with [g] [MAD]!

SC-6 Keystrokes: `g` `CLΣ` `10` `Σ+` `12` `Σ+` `14` `Σ+` `16` `Σ+` `18` `Σ+` `g` `x̄` `g` `MAD`

Answer: Mean = 14, MAD = 2.4

[g] [MAD] ([g][Σ+]) turns a 15-minute hand calculation..

SECTION 9.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Mean Absolute Deviation (MAD) — Average Distance of Data Values..

1 Big Ideas Method: 3-Step MAD Calculation

Load 2, 4, 6 into [Σ+]. Find the Mean ([g] [x-bar]) and MAD ([g] [MAD]).

Work:

Answer: _____

2 Big Ideas Method: 3-Step MAD Calculation

Load 10, 12, 14, 16, 18 into [Σ+]. Find the Mean and MAD.

Work:

Answer: _____

3 Big Ideas Method: 3-Step MAD Calculation

Load 5, 8, 8, 10, 14 into [Σ+]. Find the Mean and MAD.

Work:

Answer: _____

4 Big Ideas Method: 3-Step MAD Calculation

What is the MAD of 7, 7, 7, 7, 7 (5 identical towers)?

Work:

Answer: _____

5 Big Ideas Method: Deviation Distance from Mean

Find the Mean and MAD of: 20, 24, 28, 32, 36.

Work:

Answer: _____

6 Big Ideas Method: Deviation Distance from Mean

Find the Mean and MAD of the 6 sprint times: 12.4, 13.0, 12.6, 13.2, 12.8, 14.0.

Work:

Answer: _____

7 Big Ideas Method: Deviation Distance from Mean

Find the Mean and MAD of quiz scores: 82, 85, 88, 90, 95.

Work:

Answer: _____

8 Big Ideas Method: Deviation Distance from Mean

Team A has Mean = 50, MAD = 2.1. Team B has Mean = 50, MAD = 8.4. Which team's scores are more spread out?

Work:

Answer: _____

9 Application & 2-Way Check: Comparing MAD Consistency

For {6, 10, 11, 13}, find MAD both by averaging |x - 10| distances AND with 1 click of [g] [MAD]!

Work:

Answer: _____

10 Application & 2-Way Check: Comparing MAD Consistency

Full Chapter 9 Boss Sweep: For {5, 8, 8, 10, 14}, record [g][x-bar], [f][MED], [g][MODE], [RNG], [IQR], and [g][MAD]!

Work:

Answer: _____

BLOCK KIT: BASE-10 • Ten-Frame Zero-Pair Counters

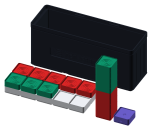
Section 10.1: Adding & Subtracting Signed Numbers — Why 'Minus a Negative' Adds!

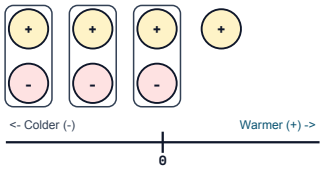
Big Ideas Method: Zero-Pair Chip Cancellation (1 Positive + 1 Negative = 0) & Add-the-Opposite Rule (a - b = a + (-b))

Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
When adding with Yellow (+1) and Red (-1) counters: if signs are the SAME, they team up (-6 + (-4) = -10); if signs are OPPOSITE, they fight in Zero Pairs (-8 + 5 = -3)! And why does subtracting a negative like 10 - (-4) equal +14? Because erasing a -4 penalty makes your score go UP by 4!

BLOCK KIT & STEP-BY-STEP GUIDE (BASE-10: TEN-FRAME ZERO-PAIR COUNTERS)





HOW TO BUILD IT WITH BASE-10 (TEN-FRAME ZERO-PAIR COUNTERS):

Step 1: Open Kit RACK-01 (Number Rack & Zero-Pair Studio) and place 5 red negative chips (-5) and 8 yellow positive chips (+8) on the tray.

Step 2: Pair up 5 red and 5 yellow chips into 5 Zero Pairs (each pair cancels to 0) and sweep them off—3 yellow chips (+3) remain!

Step 3: To subtract -4 - (-9), rewrite subtracting a negative as adding its opposite: -4 + (+9) = +5.

WORKED EXAMPLE — EXAMPLE 1: ZERO-PAIR BATTLE (-8 + 5) VS. SAME-SIGN TEAM-UP (..

Problem: Evaluate -8 + 5 and -6 + (-4) using counters and the SC-6 [+/-] key.

Big Ideas Textbook Method (Zero-Pair Chip Cancellation (1 Positive + 1 Negative = 0) & Add-the-Oppos..

1. Same Signs (+): Add the absolute values and keep the common sign: -7 + (-5) = -12.
2. Different Signs (+): Subtract the lesser absolute value from the greater and keep the sign of the greater: -5 + 8 = +3.
3. Subtracting Integers (-): Add the opposite: a - b = a + (-b), so -4 - (-9) = -4 + 9 = +5.

Block Connection (BASE-10): -8 + 5 cancels 5 Zero Pairs, leaving 3 Red (-3). -6 + (-4) combines 6 Red and 4 Red into 10 Red (-10)!

SC-6 Keystrokes: `8` `+/-` `ENTER` `5` `+` `6` `+/-` `ENTER` `4` `+/-` `+`

Answer: -8 + 5 = -3; -6 + (-4) = -10 *Always enter the number first, then tap [+/-] to make i..*

SECTION 10.1 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Zero-Pair Chip Cancellation (1 Positive + 1 Negative = 0) & Add..

1 Big Ideas Method: Zero-Pair Addition
Solve side-by-side: (a) -15 + 9 and (b) -15 + (-9).

Work:

Answer: _____

2 Big Ideas Method: Zero-Pair Addition
Solve: 14 + (-23).

Work:

Answer: _____

3 Big Ideas Method: Zero-Pair Addition
Taking Away a Penalty: Solve (a) 12 - (-8) and (b) -12 - (-8).

Work:

Answer: _____

4 Big Ideas Method: Zero-Pair Addition
Solve: -9 - 16. (Start at -9 and go 16 steps further negative!)

Work:

Answer: _____

5 Big Ideas Method: Add-the-Opposite Subtraction
Solve with decimals: -4.8 + (-3.5).

Work:

Answer: _____

6 Big Ideas Method: Add-the-Opposite Subtraction
Solve with decimals: 6.4 - 11.2.

Work:

Answer: _____

7 Big Ideas Method: Add-the-Opposite Subtraction
Solve with decimals: -7.5 - (-10.2).

Work:

Answer: _____

8 Big Ideas Method: Add-the-Opposite Subtraction
Solve with signed fractions: -3/4 - (-5/8).

Work:

Answer: _____

9 Application & 2-Way Check: Signed Decimals & Fractions
Temperature in Fairbanks is -18°F at dawn and rises to +7°F by noon. Find the change (7 - (-18)) 2 ways!

Work:

Answer: _____

10 Application & 2-Way Check: Signed Decimals & Fractions
Evaluate -2 1/2 + 4 1/4 - (-1 1/8) on the SC-6 and check by converting to eighths!

Work:

Answer: _____

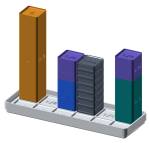
BLOCK KIT: BASE-10 • Zero-Pair & Fraction Studio

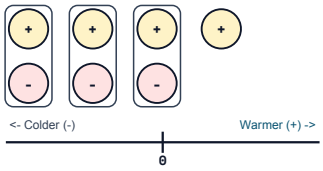
Section 10.2: Multiplying & Dividing Signed Rationals — The Light-Switch Flip Rule

Big Ideas Method: Sign Rules for Multiplying & Dividing Rational Numbers (Same Signs = +, Different Signs = -) **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:
Think of a negative sign as flipping a light switch (just like tapping [+/-]!): 1 negative flip leaves you in the dark (-), but 2 negative flips bring you right back to POSITIVE (+)! And on the SC-6, 5 [+/-] [x²] squares the entire number (-5) × (-5) = +25 without needing parentheses!

BLOCK KIT & STEP-BY-STEP GUIDE (BASE-10: ZERO-PAIR & FRACTION STUDIO)





HOW TO BUILD IT WITH BASE-10 (ZERO-PAIR & FRACTION STUDIO):

Step 1: Open Kit RACK-01 (Number Rack) and model 3 × (-4) as 3 groups of 4 red negative chips = -12.

Step 2: Treat a negative multiplier as 'Flip the Sign': (-3) × (-4) flips the 12 red chips over to 12 yellow positive chips (+12).

Step 3: Count negative factors in any product/quotient: an EVEN number of negatives gives (+), and an ODD number gives (-).

WORKED EXAMPLE — EXAMPLE 1: THE 4 SIGN COMBINATIONS & (-5)²

Problem: Compute 6 × (-4), (-6) × (-4), and (-5)² on your SC-6.

Big Ideas Textbook Method (Sign Rules for Multiplying & Dividing Rational Numbers (Same Signs = +,..

1. Same Signs -> Positive (+): (-6) × (-4) = +24 and (-24) ÷ (-4) = +6.
2. Different Signs -> Negative (-): 6 × (-4) = -24 and -24 ÷ 4 = -6.
3. Exponents with Negative Bases: (-5)² = (-5)(-5) = +25 (2 negatives = +), while (-2)³ = (-2)(-2)(-2) = -8 (3 negatives = -).

Block Connection (BASE-10): 1 negative gives -24; 2 negatives flip back to +24! For (-5)², type 5 [+/-] [x²] -> +25!

SC-6 Keystrokes: `6` `ENTER` `4` `+/-` `×` `5` `+/-` `x²`

Answer: 6×(-4) = -24, (-6)×(-4) = +2. *On algebraic calculators, typing -5² gives -25! On RP..*

SECTION 10.2 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Sign Rules for Multiplying & Dividing Rational Numbers (Same Si..

1 Big Ideas Method: Same/Different Sign Rules
Compute: (a) 7 × (-8) and (b) (-7) × (-8).

Work:

Answer: _____

2 Big Ideas Method: Same/Different Sign Rules
Compute: (a) -45 ÷ 9 and (b) -45 ÷ (-9).

Work:

Answer: _____

3 Big Ideas Method: Same/Different Sign Rules
Compute (-5)² using 5 [+/-] [x²]. Why is it +25?

Work:

Answer: _____

4 Big Ideas Method: Same/Different Sign Rules
Compute (-2)³ using 2 [+/-] [g] [x³]. Why is a negative cubed still negative?

Work:

Answer: _____

5 Big Ideas Method: Signed Powers & Rational Products
Multiply signed decimals: (-3.5) × (-2.4).

Work:

Answer: _____

6 Big Ideas Method: Signed Powers & Rational Products
Divide signed decimals: -14.4 ÷ 1.2.

Work:

Answer: _____

7 Big Ideas Method: Signed Powers & Rational Products
Multiply signed fractions: (-2/3) × (9/10).

Work:

Answer: _____

8 Big Ideas Method: Signed Powers & Rational Products
Divide signed fractions: -5/6 ÷ (-5/12).

Work:

Answer: _____

9 Application & 2-Way Check: Multi-Factor Sign Counting
Triple Product: Evaluate (-4) × (-5) × (-3) both by counting negative signs AND on your SC-6!

Work:

Answer: _____

10 Application & 2-Way Check: Multi-Factor Sign Counting
A submarine dives 360 ft in 8 minutes (-45 ft/min). Confirm the total change in 8 min and the change per minute!

Work:

Answer: _____

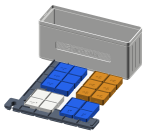
Section 10.3: Solving Two-Step Equations (ax + b = c) — Unwrapping in Reverse!

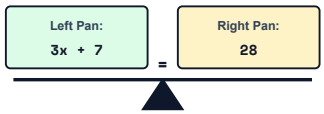
Big Ideas Method: Solving Two-Step Equations (ax + b = c) by Undoing Addition/Subtraction First, Then Multiplication/.. **Name:** _____ **Date:** _____ **Score:** ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Think of 3x + 7 = 28 like wrapping a present: first x was put into 3 boxes (× 3), then 7 loose cubes were tied on outside (+ 7). When you unwrap the present, go in REVERSE: untie the loose cubes FIRST (28 [ENTER] 7 [-] -> 21), and divide by the 3 boxes SECOND (3 [÷] -> x = 7)!

BLOCK KIT & STEP-BY-STEP GUIDE (ALGB-01: ALGEBRA BOXES & STAGE)





Step 1: 28-7=21 -> Step 2: 21÷3 = 7!

HOW TO BUILD IT WITH ALGB-01 (ALGEBRA BOXES & STAGE):

Step 1: Open Kit ALGB-01 (Algebra Boxes & Stage) and place 3 x-rods + 7 unit cubes on the Left Pan and 28 cubes on the Right Pan (3x + 7 = 28).

Step 2: Step 1 (Undo + 7): Remove 7 unit cubes from BOTH pans so the 3 x-rods equal 28 - 7 = 21 cubes (3x = 21).

Step 3: Step 2 (Undo × 3): Divide the 21 cubes equally across the 3 x-rods: 21 ÷ 3 = 7 cubes per x-rod (x = 7).

WORKED EXAMPLE — EXAMPLE 1: SOLVING 3X + 7 = 28 WITH INVERSE OPERATIONS

Problem: Solve 3x + 7 = 28 using inverse operations and check your solution.

Big Ideas Textbook Method (Solving Two-Step Equations (ax + b = c) by Undoing Addition/Subtraction..

1. Step 1 (Undo Addition or Subtraction): Subtract 7 from both sides of 3x + 7 = 28 -> 3x = 21.
2. Step 2 (Undo Multiplication or Division): Divide both sides by 3 -> x = 21 ÷ 3 = 7.
3. Check by substituting x = 7 back into the original equation: 3(7) + 7 = 21 + 7 = 28.

Block Connection (ALGB-01): Step 1 (Clear loose cubes): 28 - 7 = 21 (so 3x = 21). Step 2 (Split across rods): 21 ÷ 3 = 7! Check: 3(7) + 7 = 28!

SC-6 Keystrokes: 28 | ENTER | 7 | - | 3 | ÷

Answer: x = 7 (Step 1: 3x = 21) *Step 1: Subtract 7 (3x = 21). Step 2: Divide by 3 (x =..*

SECTION 10.3 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Solving Two-Step Equations (ax + b = c) by Undoing Addition/Sub..

1 Big Ideas Method: 2-Step Inverse Unwrapping

Solve 4x + 9 = 45. Write the Step 1 value (4x = ?) and final x.

Work:

Answer: _____

2 Big Ideas Method: 2-Step Inverse Unwrapping

Solve 5x - 12 = 38. (Undo -12 with [+] first, then ÷ 5!)

Work:

Answer: _____

3 Big Ideas Method: 2-Step Inverse Unwrapping

Solve 6y + 11 = 59.

Work:

Answer: _____

4 Big Ideas Method: 2-Step Inverse Unwrapping

Solve 8n - 15 = 49.

Work:

Answer: _____

5 Big Ideas Method: Fractional & Negative Coefficients

Solve m / 4 + 6 = 15. (Subtract 6 first, then multiply by 4!)

Work:

Answer: _____

6 Big Ideas Method: Fractional & Negative Coefficients

Solve k / 3 - 5 = 7.

Work:

Answer: _____

7 Big Ideas Method: Fractional & Negative Coefficients

Solve -3x + 14 = 2.

Work:

Answer: _____

8 Big Ideas Method: Fractional & Negative Coefficients

Solve 2.5w + 8 = 23.

Work:

Answer: _____

9 Application & 2-Way Check: 2-Step Word Problem Modeling

A repair tech charges $35 flat plus $25/hr (25h + 35 = 160). Find hours h by unwrapping AND plug-in check!

Work:

Answer: _____

10 Application & 2-Way Check: 2-Step Word Problem Modeling

You save $15/week toward a $120 skateboard and already have $45 (15w + 45 = 120). How many weeks w?

Work:

Answer: _____

Section 10.4: Probability, Combinatorics & Graduation Quiz Sprint ([DICE], [x!], [f] [nCr], [f] [?])

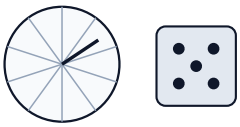
Big Ideas Method: Circle Circumference (C = πd = 2πr) & Area (A = πr²), Theoretical Probability & Counting (n!, nPr, nC.. Name: _____ Date: _____ Score: ____ / 10

CONCEPT SUMMARY & TEXTBOOK CONNECTION:

Probability is Favorable Outcomes ÷ Total Outcomes—and you can express it as a Fraction, Decimal, or Percent! Plus, the top-left keys of your SC-6 unlock Combinatorics: [x!] ([g][DICE]) for arrangements of all n items, [g] [nPr] when order matters (1st/2nd/3rd podium), [f] [nCr] when order does NOT matter (pizza toppings), and [f] [?] ([f][C]) for the built-in Math Quiz Engine!

BLOCK KIT & STEP-BY-STEP GUIDE (PROB-01: PROBABILITY DICE & SPINNER STUD..





[DICE] • [x!] • [f][nCr] • [g][nPr]

HOW TO BUILD IT WITH PROB-01 (PROBABILITY DICE & SPINNER STUDIO):

Step 1: Open Kit PROB-01 (Probability & Circle Studio) and roll the circular disk of diameter d along the ruler—1 turn covers π × d (≈ 3.14 × d).

Step 2: Spin the 10-sector decagon spinner and roll the 6-sided die to count favorable outcomes vs. total outcomes.

Step 3: Arrange 4 colored blocks in order to see 4! = 4 × 3 × 2 × 1 = 24 permutations.

WORKED EXAMPLE — EXAMPLE 1: 3-WAY PROBABILITY & FACTORIAL ([X!])

Problem: A 10-sector spinner has 4 Blue, 3 Gold, and 3 Red sectors. Find P(Blue) in 3 forms, and find how many ways to line up 5 trophies (5!).

Big Ideas Textbook Method (Circle Circumference (C = πd = 2πr) & Area (A = πr²), Theoretical Probabil..

1. Circle Formulas (with r = 5, d = 10): Circumference C = πd = 10π ≈ 31.42; Area A = πr² = 25π ≈ 78.54.
2. Theoretical Probability: P(event) = Favorable Outcomes / Total Outcomes, and P(not E) = 1 - P(E).
3. Counting Principle: Permutations (order matters) use n! or nPr; Combinations (order doesn't matter) use nCr.

Block Connection (PROB-01): P(Blue) = 4/10 = 2/5 = 0.40 = 40%. For 5 trophies: press 5 [g] [x!] ([g][DICE]) -> 120 ways!

SC-6 Keystrokes: `.` `4` `.` `10` `g` `SIMP` `5` `g` `x!`

Answer: P(Blue) = 2/5 = 0.4 = 40%; 5..

5! means 5 × 4 × 3 × 2 × 1 = 120!

SECTION 10.4 PRACTICE — WORK OUT EACH PROBLEM & WRITE YOUR ANSWER

Method Focus: Circle Circumference (C = πd = 2πr) & Area (A = πr²), Theoretical Probability & Counting (n!,..

1 Big Ideas Method: Circle Circumference & Area (π)

Roll a fair 6-sided die. Find P(rolling a 5) as a fraction, and P(rolling an even number) in simplest form.

Work:

Answer: _____

2 Big Ideas Method: Circle Circumference & Area (π)

A 10-sector spinner has 4 Blue, 3 Gold, 3 Red. Find P(Blue) as a simplified fraction, decimal, and percent.

Work:

Answer: _____

3 Big Ideas Method: Circle Circumference & Area (π)

Using the same 10-sector spinner, find P(NOT Red) as a fraction and percent.

Work:

Answer: _____

4 Big Ideas Method: Circle Circumference & Area (π)

In how many ways can 4 students line up at the door? Compute 4! using [g] [x!].

Work:

Answer: _____

5 Big Ideas Method: Theoretical Probability & Complement

In how many ways can you arrange 6 books on a shelf? Compute 6! using [g] [x!].

Work:

Answer: _____

6 Big Ideas Method: Theoretical Probability & Complement

How many ways can you choose a 3-person team from 8 players (order does NOT matter: [f] [nCr])?

Work:

Answer: _____

7 Big Ideas Method: Theoretical Probability & Complement

How many ways can 1st, 2nd, and 3rd place be awarded among 8 finalists (order DOES matter: [g] [nPr])?

Work:

Answer: _____

8 Big Ideas Method: Theoretical Probability & Complement

Two Independent Events: You flip a coin (1/2) AND roll a 6-sided die (P(>4) = 2/6 = 1/3). Multiply to find P(Heads and >4)!

Work:

Answer: _____

9 Application & 2-Way Check: Permutations (nPr) & Combinations (nCr)

From 10 toppings, compare choosing 3 toppings ([f] [nCr]) vs. 3 ordered medals ([g] [nPr]). How are they related by 3!?

Work:

Answer: _____

10 Application & 2-Way Check: Permutations (nPr) & Combinations (nCr)

GRADUATION BOSS: Compute 5! both with [g] [x!] AND 5 [ENTER] 5 [g] [nPr], then complete a 5-streak on [f] [?]!

Work:

Answer: _____